

Payments & fintech

SURAT, 14 OCTOBER 2026, 9 PM

Hitesh Shah (an illustrative persona) sells sarees wholesale from a shop in Surat's textile market. Retailers from across Gujarat pay him on a PhonePe QR code taped to the counter, usually ₹20,000 to ₹60,000 at a time, about ₹25 lakh a month. For ten years UPI has cost him nothing. Tonight a WhatsApp forward from his trade association says that from tomorrow, merchants like him pay a fee: 0.4% on every UPI payment above ₹2,000.

He does the maths on a calculator: ₹25 lakh × 0.4% = **₹10,000 a month**, ₹1.2 lakh a year. His first idea is the one thousands of traders will have: "I'll ask buyers to pay my personal UPI ID instead. Person-to-person transfers are still free."

Try it: you are the PM for PhonePe's merchant app. What changes for Hitesh on 15 October, and what would you build? Take five minutes before reading on.

How money moves in India, and who gets paid

7:40 pm, a kirana in Pune. Asha buys ₹500 of groceries, scans the QR code at the counter and types her UPI PIN. Before she has put her phone away, the shop's soundbox says "₹500 received". PhonePe was on her screen the whole time, yet PhonePe never held her ₹500 and earned nothing from the payment. It is still one of India's most valuable fintech companies.

24.51 bn UPI payments in Aug 2026, up 22% on Aug 2025	791 mn payments a day, about 9,150 every second on average	₹29.82 lakh Cr moved over UPI in Aug 2026: average ₹1,217 a payment	12.3 crore credit cards in India (Jul 2026), against ~59 crore Jan Dhan accounts
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Five rails, one country

A **payment rail** is the shared road that money messages travel on, like a railway line many trains can use. India has five that matter to a PM. They differ in who runs them, how big a payment they suit, and above all **who pays for each payment**, which decides what products can be built on top.

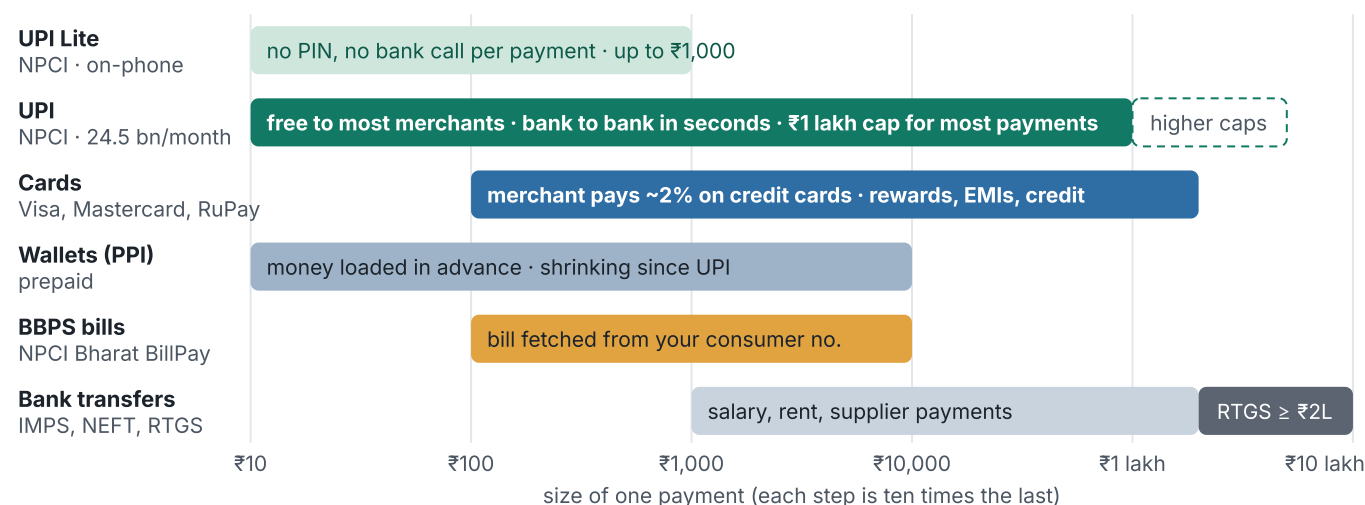
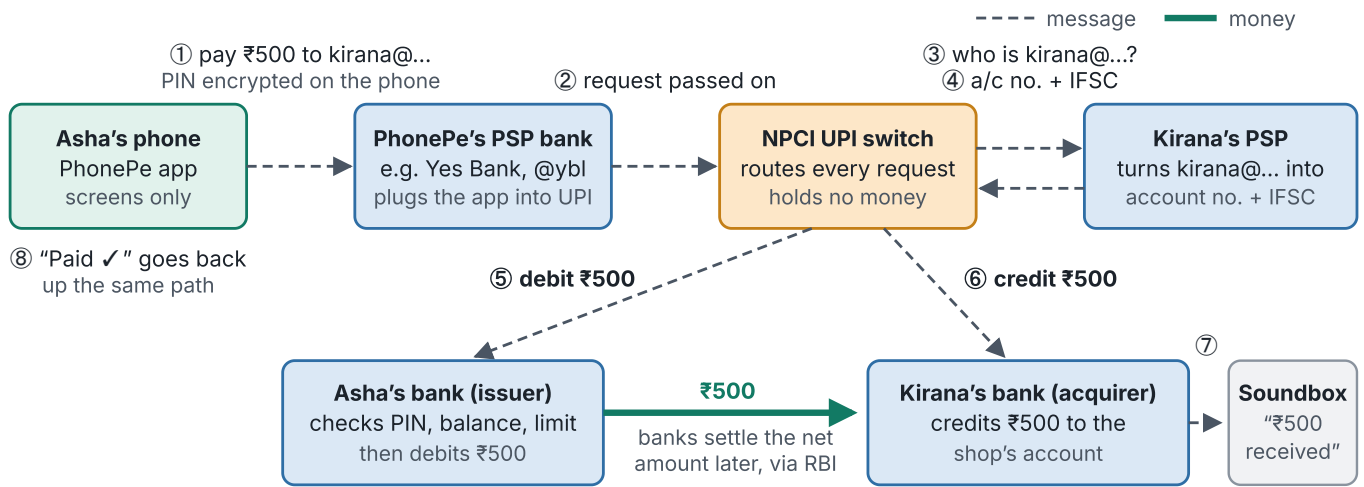


Figure 1.1 · Which rail carries which payment. Bars show the payment sizes each rail usually carries. UPI now covers almost the whole everyday range and is free to most merchants, so it has taken volume from wallets and debit cards. Cards survive on credit and rewards, which the ~2% merchant fee pays for. NPCI and RBI product rules (UPI Lite ₹1,000 per payment from Dec 2024; RTGS minimum ₹2 lakh; higher UPI caps for tax, hospitals, education and investments); UPI volume Aug 2026. Bar extents are typical ranges, not legal limits.

How one UPI payment moves

Follow Asha's ₹500. Four organisations pass messages before any money moves, and the app she looks at is only the first of them.



Money never sits in PhonePe: the app only carries messages; banks move the money.

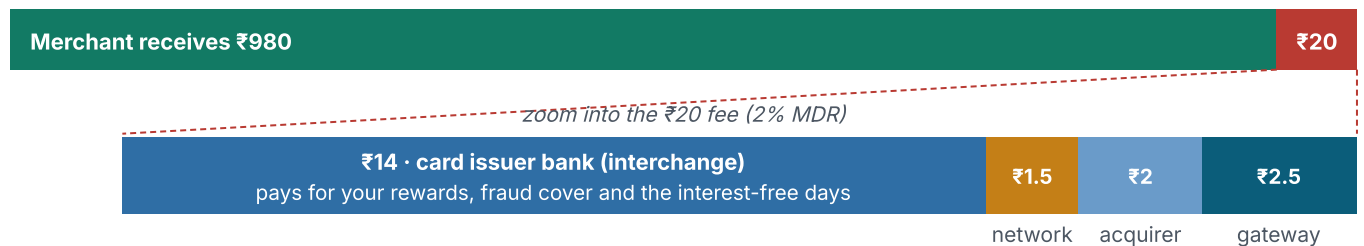
Figure 1.2 · One ₹500 UPI payment, step by step. Follow the dashed arrows: four organisations pass messages before any money moves, and the only real money movement (green) is bank to bank. The app owns the screen and the customer relationship, but not the rails. Flow simplified from NPCI's UPI procedural guidelines; PSP handle as used by PhonePe, Sep 2026.

UPI ID (or VPA, virtual payment address) is a nickname for a bank account, such as kirana@ybl. A **TPAP** (third-party app provider) such as PhonePe or Google Pay builds the screens but cannot join UPI alone; it plugs in through one or more **PSP banks** (payment service providers). **NPCI**, a not-for-profit owned by banks, runs the switch. The **issuer** is the payer's bank and the **acquirer** is the merchant's. Because every UPI app talks to every bank, any app can pay any QR code. That single fact, called **interoperability**, makes switching apps free for users and explains why UPI apps fight so hard for habit.

Who pays for a "free" payment?

Every payment has costs: servers, fraud checks, customer support, bank systems. On a card, the merchant pays for all of it through the **MDR** (merchant discount rate: the share of each payment a merchant pays to accept it). The MDR is split along the chain, and the biggest slice, the **interchange**, goes to the bank that issued the card.

A ₹1,000 credit-card payment at a clothes shop



The same ₹1,000 by UPI at a small shop



Figure 1.3 · Where a card fee goes, and why UPI has no rewards. The issuer's ₹14 is what funds your card's 1% cashback. On UPI the same slot is ₹0, so any cashback an app gives comes out of its own marketing budget. Typical split, illustrative: actual interchange varies by card type, merchant category and contract (industry estimates, 2023–25). Zero UPI MDR: in force since Jan 2020.

Since January 2020 merchants have paid zero MDR on UPI. The government paid banks a small subsidy instead (₹1,500 Cr in FY25, at 0.15% on small-merchant payments up to ₹2,000). Zero MDR is why a chaiwala takes UPI and not cards. It is also why UPI apps lose money on payments. That changes on **15 October 2026**, when large merchants start paying 0.4% on UPI payments above ₹2,000.

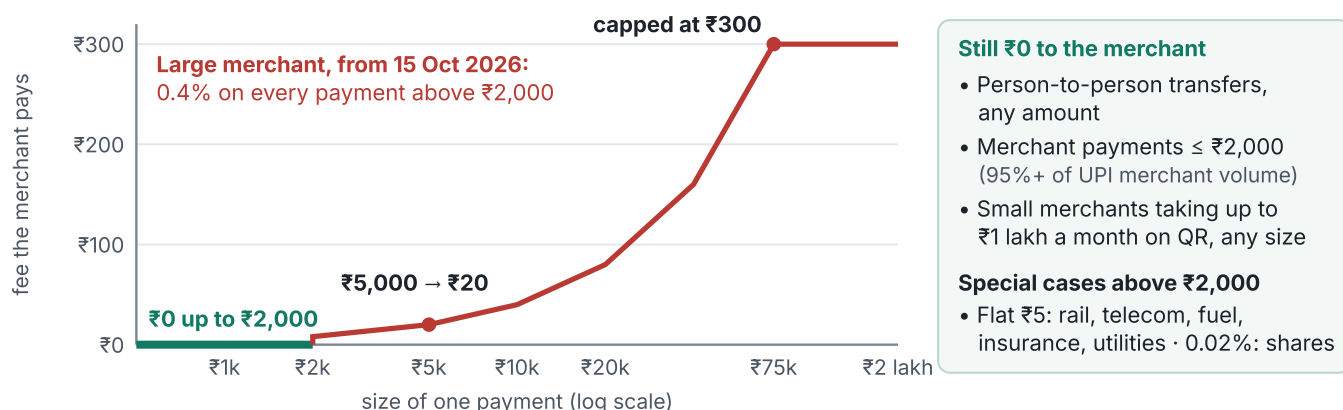


Figure 1.4 · What a merchant pays to accept UPI from 15 Oct 2026. The fee hits only large merchants and only above ₹2,000, so most everyday payments stay free. How the 0.4% is split between banks, the app and NPCI has not been published, and that split decides whether PhonePe's payments business ever makes money. *NPCI MDR FAQ, 15 Sep 2026 (via SCC Online, Business Standard); PIB, FY25 incentive scheme.*

How a free app makes money: follow one loan

If payments earn nothing, the habit must be sold. The biggest thing it sells is credit. The app becomes a **lending service provider** (LSP): it finds the borrower, shows the offer and handles the screens, while a bank or NBFC (a non-bank lender licensed by RBI) lends its own money and carries the risk. Here is one ₹1 lakh personal loan, followed rupee by rupee.

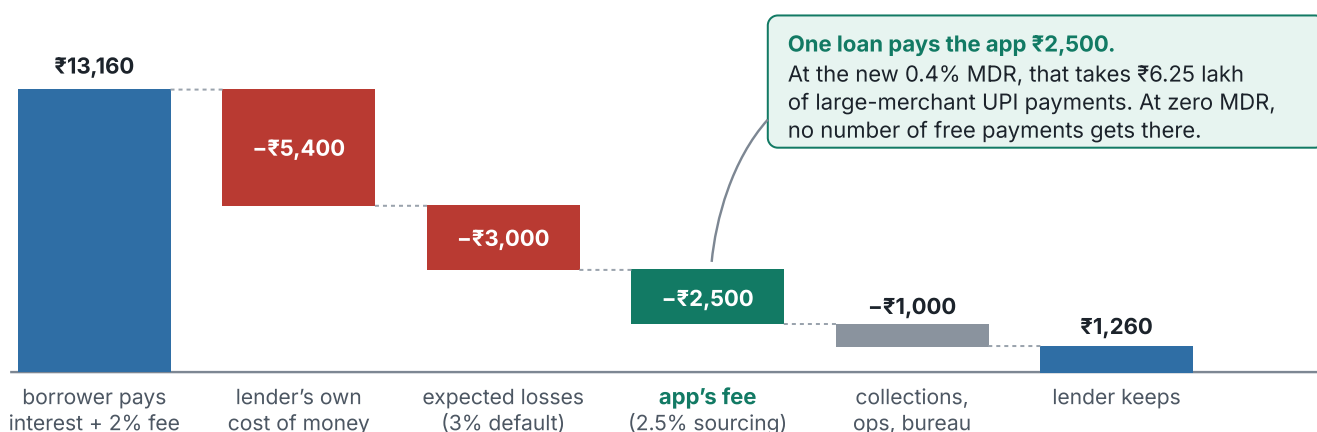


Figure 1.5 · Where the money goes on a ₹1 lakh, 12-month app loan at 20% a year. The lender funds the loan and takes the losses; the app earns a fee for finding and serving the borrower. Watch the red bars: if defaults rise from 3% to 6%, the lender's ₹1,260 turns into a loss, which is why lenders cut app partners fast when credit quality slips. *Illustrative numbers; EMI arithmetic exact (EMI ₹9,263; interest ₹11,160). Rules: RBI (Digital Lending) Directions, 8 May 2025.*

Two RBI rules shape every lending feature you will design. First, the loan must go **straight into the borrower's bank account**, never through the app's or a third party's account. Second, an app may promise to cover the lender's losses, a **default loss guarantee** (DLG), but only up to **5% of the loans** it sourced. A DLG earns the app a bigger fee and also makes it share the pain when borrowers default.

Players and numbers

Player	What it is	Latest numbers (dated)
PhonePe	Largest UPI app (Walmart-owned); payments plus lending, insurance and broking	FY26 revenue ₹7,920 Cr (+11%), loss ₹2,792 Cr. IPO approved by SEBI Jan 2026, paused Mar 2026
Google Pay	Google's UPI app; people-first design, rewards, loans via partners	Indian entity's revenue ₹1,547 Cr in FY25, profit ₹50 Cr. Launched a RuPay credit card with Axis Bank in Dec 2025
Paytm	Merchant payments (soundboxes, card machines) plus loan and insurance distribution	FY26 revenue ₹8,437 Cr, first full-year profit ₹552 Cr. Payments Bank licence cancelled Apr 2026
Navi, super.money, CRED	Challengers using cashback, credit on UPI and card-bill payments	Navi 4.4% of UPI (Aug 2026), revenue ₹2,565 Cr and a ₹126 Cr loss in FY25

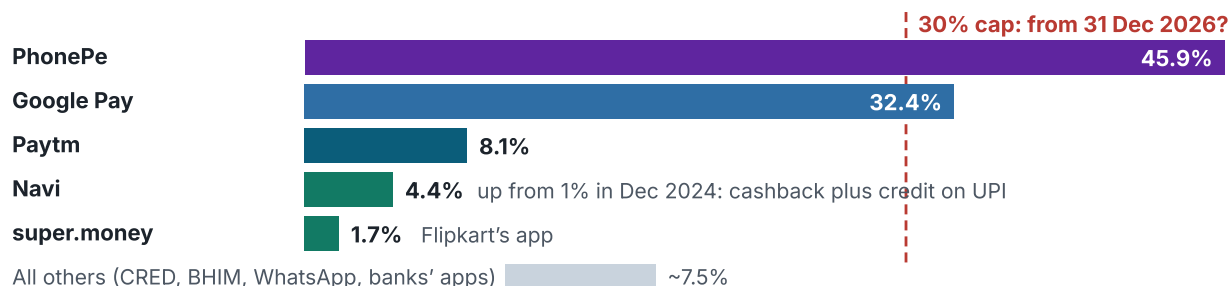


Figure 1.6 · Share of UPI payments by app, August 2026. Two apps start almost four in five UPI payments. If NPCI enforces its 30% cap, neither leader could add new users, and the growth would flow to the apps in green. *NPCI app-wise data via Inc42, Aug 2026; cap timeline: NPCI extensions, latest to 31 Dec 2026.*

Turning points



Figure 1.7 · Ten years of UPI in nine moments. Green dots are growth, amber dots are rules, hollow red dots are still to come. The rules shaped the market more than any product launch: zero MDR decided who could make money, and the cap decides who may keep growing. *NPCI; Business Standard (Aug 2023, Aug 2025); Business Today (cap extension, 31 Dec 2024); RBI press releases, 31 Jan 2024 and 24 Apr 2026.*

Demonetisation (Nov 2016) created the users. With 86% of cash by value withdrawn overnight, people who had never paid digitally had to. Paytm's wallet grew first, because UPI apps were still clunky; UPI caught up within two years because it moved bank money directly, with no wallet to top up.

Zero MDR (Jan 2020) decided the business model. Once merchants paid nothing, no app could earn from payments, so the winners became the apps that could sell something else to their users: loans, insurance, investments, soundbox rentals.

Paytm's bank (Jan 2024 to Apr 2026) showed the regulatory risk. RBI stopped Paytm Payments Bank taking deposits after repeated compliance failures, and cancelled its licence in April 2026. Paytm survived by moving its users' UPI IDs to four partner banks, a lesson in why PMs in fintech sit close to compliance teams.

The metrics that matter

Metric	What it means	Worked number, and why it matters
Volume and TPV	Count of payments, and total payment value (₹ moved)	Aug 2026: ₹29.82 lakh Cr ÷ 24.51 bn = ₹1,217 average ticket . A falling ticket means UPI is winning small, everyday payments.
Market share	Share of all UPI payments started in one app, by count	PhonePe 45.9%. The 30% cap is measured on this, over a rolling three months.
Success rate and TD	Share of payments that go through. Technical decline (TD): a failure caused by a bank or system, not the user	Each 0.1 point of TD at 791 mn a day is about 7.9 lakh failed payments daily. Users blame the app they see.
MTU	Monthly transacting users: at least one successful payment in 30 days	Better than downloads or registrations (PhonePe: 70 crore+ registered), because it counts habit.
Take rate	Revenue ÷ TPV	About 0% on most UPI payments; 2% on a credit card; 0.4% on eligible large-merchant UPI from Oct 2026.
Approval rate and 30+ DPD	Share of loan applicants approved; share of loans 30 or more days past due	If the app shows 100 loan offers and 20 users apply, 8 get approved (40%), and 2 of every 100 loans go 30+ days overdue. Chase approvals and the second number rises.
Fraud rate	Fraud losses per ₹ moved, often in basis points (1 bp = 0.01%)	₹981 Cr of UPI fraud in FY25 on roughly ₹260 lakh Cr moved is under 0.4 bp, small in aggregate but devastating to each victim.

The non-obvious mechanics

Mechanic	How it works	Why a PM cares
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The 30% cap	An app above 30% of UPI volume would have to stop adding <i>new</i> users. Announced Nov 2020, deferred three times, now due 31 Dec 2026	Growth plans at PhonePe and Google Pay hang on it; challengers plan around it
PSP-bank model	Apps join UPI through partner banks and spread users across handles (@ybl, @ibl, @axl)	A failing partner bank can be routed around; Paytm used this to survive its bank's closure
Credit on UPI	RuPay credit cards linked to UPI (since 2022) and pre-approved bank credit lines on UPI (since 2023)	Merchants pay card-like fees above ₹2,000 on these, so credit is the first real money on UPI
UPI Circle	A primary user lets up to five people pay from their account: full delegation up to ₹15,000 a month (₹5,000 a payment) or approval of each payment	Payments for teenagers and elderly parents without a new bank account
UPI Lite and 123Pay	Lite: small payments up to ₹1,000 from an on-phone balance, no PIN. 123Pay: UPI on feature phones by voice call or missed call, up to ₹10,000	The existing answers to "bad network" and "no smartphone"
Failed-payment rule	If a failed UPI payment debited the customer, the bank must reverse it by the next working day or pay ₹100 a day	"Pending" screens and refunds are a product surface, not only an operations task

CHECK · "PHONEPE MUST CUT ITS SHARE TO 30% BY JANUARY 2027" **OVERSTATED**

The rule says an app above 30% of UPI volume (rolling three months) must stop *onboarding new users*. It does not force existing users off. NPCI has pushed the deadline back three times, most recently to 31 December 2026, and as of September 2026 it is not in force. Watch it anyway: a freeze on new users would hand growth to Paytm, Navi and CRED.

IN THE INTERVIEW · FIVE QUESTIONS THIS SECTOR LOVES

1. How does PhonePe make money if UPI is free? *The habit is distribution for loans, insurance and bills.*
2. Design a payments product for rural users, teenagers or the elderly. *Start from who holds the bank account and the phone.*
3. Payment failures went up. Why? *Split by bank and by step before guessing.*
4. How would you reduce fraud without hurting good users? *Name the safety check you will watch.*
5. What happens to UPI apps when MDR returns or the 30% cap bites? *Who pays, who switches, who grows.*

REAL-LIFE EXAMPLE · WHAT 0.4% MEANS FOR THREE SHOPS

Shop (illustrative)	UPI a month	Above ₹2,000	Fee a month
Kirana, Pune	₹90,000	₹5,000	₹0 (small merchant)
Phone shop, Nagpur	₹8 lakh	₹3.2 lakh	₹1,280
Hitesh, Surat	₹25 lakh	₹25 lakh	₹10,000

The fee lands on a few high-ticket merchants. They are exactly the ones who can ask buyers to pay a personal UPI ID, or switch to NEFT, to dodge it.

IN THE INTERVIEW · "HOW DOES PHONEPE MAKE MONEY?" IN 60 SECONDS

"Almost nothing from the payment itself. UPI has been free to merchants since 2020, so the 45.9% share is a distribution asset, not a revenue line. Money comes from what the habit sells: its lending arm earned ₹945 Cr in FY26, plus insurance, broking, bill payments and soundbox rentals. Two things could change the maths: the 0.4% fee on large merchants from 15 October, if PhonePe gets a fair slice of it, and the 30% cap, which would stop it adding users. I'd watch revenue per monthly transacting user, not payment volume."

CASE 1.1 · A BANK APP FOR RURAL INDIA

Product design Hard IIM B Sigma PM Casebook 2026–27, pp. 55–56

Try it first. Read only the question in the blue box. Take 5 minutes and say your structure out loud before you read the candidate's answer.

THE QUESTION

You're a Product Manager for X Bank in India. Design a mobile application catering to Rural India.

A Thank you very much for the case. Before we move forward, I have a few clarifying questions: Do we assume that the bank already has KYC-authenticated customers in rural India and we want to enable e-services? In which geographies does our bank operate? How does the consumer currently use services of our bank/ other banks? Is there a particular objective behind launching the app?

Q You may assume that the bank already has a vast rural customer base across India. Currently the bank offers its services only through its branches, however due to lack of infrastructure in many rural areas, the branches are often overcrowded. This is precisely what the bank wants to fix- enhance user experience while availing Banking Services.

A Customers availing services can be broadly classified in 2 types:

- Vendors
- Regular Customers

Is there a particular category of users you would like me to explore?

Q You're free to choose and proceed.

A As our primary goal is to improve the User Experience while availing the banking services, I would like to create the app for regular customers due to the sheer volume of users. We can later look into expanding the idea for vendors. Some of the primary issues faced by a regular customers while visiting the branch would be:

- Long queues for depositing money- Due to lack of money depositing ATMs, a lot of customers would have to visit the branch.
- UPI transaction failure rate-Due to poor network coverage, rural India faces ~30% transaction failure rates. This may lead of failed critical payments-thus mandating a visit to the branch.
- Govt. Scheme Money Withdrawals- Customers may not be well aware of all the schemes they may be eligible for and documents required to apply. This may lead to multiple enquiries and check-ins related to the money deposit schedule etc.

These are all the issues, I could think of. Am I missing anything or is there a particular issue faced by the customers which you would want me to explore?

Q This seems to be apt. Please proceed with your analysis.

A Next, I would like to explore some features to address the pain points we discussed earlier. Also, we should note that the app should be made as simple and light-weight as possible keeping in mind the types of smartphones most of rural India may use.

- Vendor Partnership-Most transactions in rural India is still cash-based. Vendors regularly visits the bank to deposit their revenue. Hence, the vendor partners can collect cash on behalf of regular customers and deposit in bank on their next visit. The app can display the nearest vendor, and the transaction will be recorded for regular customers as soon as both party approve that the cash has been transferred.
- Offline UPI- NPCI already has an SMS-based UPI feature, however the feature lacks any UI and requires customer to remember the number which is used to interact etc. This app feature can work offline, with the SMS read/ write permissions and the transaction can be complemented with a UI.
- Scheme Eligibility- Based on all the KYC data and banking information, the app can display all the schemes the customer may be eligible for, a way to apply from anywhere by scanning documents/ listing all the documents to bring to apply

Q Let's say you're planning to launch the features in a phased manner; how will you prioritize?

A To prioritize the features discussed, I would like to look at 4 different parameters, Expected reach, impact, confidence & effort of the features.

Feature	Reach	Impact	Confidence	Effort	Priority
Vendor Partnership	High-Moderate	High-Moderate	Moderate-Low	High	3
Offline UPI	High	High	High	Moderate	1
Scheme Eligibility	High-Moderate	High	High	Low	2

A Basis on my understanding and judgement, I would like to develop and launch the MVP of the app with Offline UPI first, followed by Scheme Eligibility and Vendors Partnerships.

Q Which metrics would you monitor to measure the success?

A To monitor the success of our metrics, we can have different levels of metrics.

- L1 metrics- Average waiting time of a customer during bank visit
- L2 metrics-
 - DAU/MAU, Number of App downloads, number of application crashes,
 - Number of Vendors onboarded/ Vendor Transactions, fault rate (no. of disputed transactions with the vendors),
 - Number of offline UPI transactions,
 - Number of customers checking eligibility for govt. schemes.

Q Thank you, we can close the discussion for this case here.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked for the objective.** The answer, “branches are over-crowded”, turned a vague “design an app” into a sharp brief: take load off branches.
- ✓ **Asked how customers bank today** before proposing anything.
- ✓ **Picked one segment with a reason** (regular customers, for volume) and parked vendors for later.
- ✓ **Prioritised with named criteria**, proposed an MVP order, and made the top metric (waiting time at the branch) match the goal.
- ✗ **Never asked about phones, network or language.** In rural India many people share a phone: 18% of India’s internet users go online on someone else’s device, and about 80% of them are rural (IAMAI-Kantar 2025).
- ✗ **Scored RICE in words** (“High-Moderate”), so the ranking is still a gut call. With rough numbers the order changes (Figure 1.10).
- ✗ **Reinvented what exists.** “Vendors collecting cash” is the business correspondent model: 17.36 lakh bank agents already do it (Jul 2026). Offline UPI exists as UPI 123Pay.
- ✗ **Metrics mixed goal and vanity.** Downloads and DAU/MAU don’t show fewer branch trips, and nothing watches for fraud through agents.

CHECK · “RURAL INDIA FACES ~30% UPI TRANSACTION FAILURE RATES” DOESN’T HOLD UP

No public source gives a rural figure near 30%. Nationally, technical declines (failures caused by banks or systems) fell from 8–10% in UPI’s first year to under 1% by 2025, and NPCI’s target is below 0.25%. Weak signal does cause timeouts and “pending” payments, and those feel like failures. The honest version: “Poor network makes payments slow or pending more often in villages; I don’t have a number, so I’d check our own data by district.”

CHECK · “OFFLINE UPI IN OUR APP, USING SMS READ/WRITE PERMISSIONS” DOESN’T HOLD UP

Google Play allows the SMS read and send permissions only for the phone’s default SMS app and a short list of approved uses, so a bank app would struggle to get them. The need is real, and it is already met: UPI 123Pay lets feature-phone users pay by voice call or missed call, up to ₹10,000 a payment. The better product move is to promote 123Pay in the bank’s alerts, not to rebuild it.

THE SCOPING KIT · FIVE QUESTIONS BEFORE ANY ANSWER

A good doctor doesn’t write a prescription the moment you say “headache”. They ask where it hurts, since when, and whether anything changed: new food, new medicine, less sleep. The scoping kit is your version of those questions. It takes about a minute and stops you solving the wrong problem. In this case, question 2 alone changed the brief from “an app” to “emptier branches”.

Same questions, everyday setting: planning a friend’s birthday, you’d ask what they’d enjoy (goal), who is coming (who), what’s already booked (today), and the budget and date (limits).

1

What exactly is it?

So you and the interviewer mean the same thing.

“A mobile app for X Bank’s existing rural customers. Right?”

✗ **half: asked about KYC, not the product**

2

What’s the goal?

Growth, usage, money, cost or trust lead to different answers.

“Is there a particular objective behind the app?”

✓ **asked: the key question here**

3

Who is it for?

Farmers, labourers and shopkeepers bank differently.

“In which geographies does our bank operate?”

✓ **asked, then chose a segment**

4

What happens today?

Shows the real pain and why now.

“How does the consumer currently use our services?”

✓ **asked**

5

What are the limits?

Phones, network, language and rules shrink the design.

“What phones do they have? Which languages?”

✗ **skipped**

For a design case, add one or two: which journey stage? · which platform first (smartphone, feature phone, both)? · any legal limits (who may handle cash)? · say the success metric at the end.

Rule of thumb: don’t ask everything. Ask 3–4, say one assumption out loud (“I’ll assume about half our rural customers have a smartphone”), and begin.

STRONGER ANSWER



Figure 1.8 • A design answer in six moves. The same order works for any “design X for Y” case. Each turn below is tagged with its move. *Author’s summary.*

A ① CLARIFY

Three quick questions. What’s the goal: fewer branch visits, more digital use, or lower cost? Who are our rural customers, and what phones do they carry? Any languages or rules I should design around?

Q Goal: emptier branches. Farmers, labourers, small shopkeepers. About half have a smartphone, often shared. Pick the languages you like.

Why this move: “emptier branches” is now the test for every idea. A feature that doesn’t remove a trip is out, however clever.

A ② PICK USER • ③ FIND THE PAIN

I’ll design for the account holder who visits most. I’ll assume 1 crore rural customers. A trip costs them a morning: 8 km each way, a two-hour queue, about ₹200 of lost wages. And many trips only ask “has my money come?”, a question the bank already knows the answer to (Figure 1.9).

Why this move: sorting visits by reason shows which ones an app can remove.

A ④ IDEAS

One idea per reason for visiting. **Money-arrived alerts:** a voice message in the local language when wages or scheme money land, plus a missed-call balance check. **Scheme checker:** which schemes you qualify for, and the documents to bring. **Branch token:** book a time slot. **Agent finder:** the nearest bank agent for cash deposits. Every feature needs a voice or missed-call version for the half without a smartphone.

Q Why not the offline UPI idea?

A UPI 123Pay already works on feature phones, and Android won’t let a bank app send SMS in the background. I’d promote 123Pay inside our alerts instead of rebuilding it.

A ⑤ PRIORITISE

Scoring with RICE, per quarter, with rough numbers: alerts reach 40 lakh people, impact 1, confidence 80%, effort 2 person-months: $40 \times 1 \times 0.8 \div 2 = 16$. The agent finder reaches more people but needs agent training and cash limits, so effort is 8 and it scores under 4. Alerts first, then the scheme checker and tokens; agents in phase 2.

Why this move: numbers, even rough ones, let the interviewer argue with your inputs instead of your instinct.

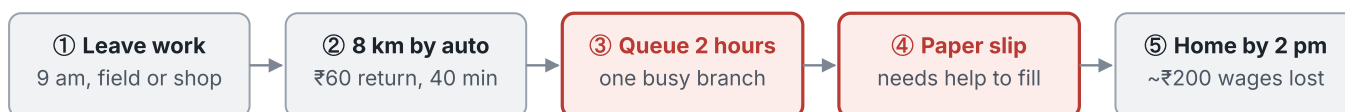
A ⑥ MEASURE

Goal (North Star): branch visits per rural customer per month, from about 1.5 to 1.1 in six months (illustrative).

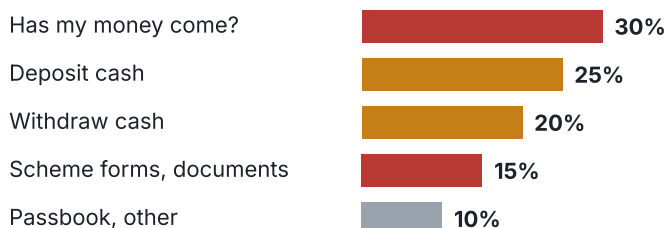
Safety check (guardrail): fake “money arrived” scam messages and complaints about missed alerts. **Trade-off:** messages cost money: 40 lakh people $\times$ 4 alerts $\times$ ₹0.15 = ₹24 lakh a month, far less than the branch staff time saved.

Next: pilot in two districts before a state-wide launch.

A cash deposit or balance check today: one morning, five steps



Why people came (share of visits, illustrative)



What could replace the trip

voice alert, missed-call balance
 nearest bank agent (BC)
 agent, micro-ATM
scheme checker + document list
 branch token for the rest

Figure 1.9 • A rural branch trip, and the reasons behind it. Red rows are trips an app can remove completely, because the bank already has the answer. Amber rows still need cash in someone’s hand, so they need an agent, not only an app. *Journey and visit shares are illustrative, for teaching; agent network: 17.36 lakh BCs and 99.92% of villages within 5 km of a banking outlet, govt data, Jul 2026.*

FRAMEWORK • RICE: RANK IDEAS WITH FOUR ROUGH NUMBERS

1 • The situation. The candidate had three good ideas and had to pick an order. Without numbers, “High-Moderate” against “High” is a matter of taste, and the interviewer can only nod or disagree.

2 • An everyday example. A free Saturday and four jobs at home: fix the leaking tap, paint the balcony, set up the family’s new phone, clean the fan. You naturally ask: how many people does it help, how much, how sure am I it’ll work, and how long will it take?

RICE part	The question	Saturday jobs	Rural bank app
Reach	How many people, in a set time?	Tap: all 4 at home. Phone: 1 (grandma)	Alerts: 40 lakh customers a quarter
Impact	How much for each? (3 huge, 2 high, 1 medium, 0.5 low)	Tap: 1, saves water. Phone: 3, she can video-call	Alerts: 1, removes a trip for some
Confidence	How sure are the numbers? (100%, 80%, 50%)	Tap: 80%, done it before. Paint: 50%	Alerts: 80%, simple tech; agents: 50%
Effort	How much work? (person-months)	Tap: 1 hour. Balcony: 2 days	Alerts: 2; agent finder: 8

3 • The framework. **Score = Reach × Impact × Confidence ÷ Effort.** A high score means a lot of benefit per unit of work. Popularised by the product team at Intercom, it works because each part is argued separately: someone can challenge your reach without rejecting the whole idea.

RICE scores, per quarter (illustrative inputs; bar = score)

$R (\text{lakh}) \times I \times C \div E$

Money-arrived alerts		16.0	$40 \times 1 \times 0.8 \div 2$
Scheme checker		8.3	$25 \times 2 \times 0.5 \div 3 \cdot \text{candidate's \#2}$
Branch token		8.0	$20 \times 1 \times 0.8 \div 2$
Agent finder + deposits		3.8	$30 \times 2 \times 0.5 \div 8 \cdot \text{candidate's \#3}$
Offline UPI by SMS		0.8	$15 \times 1 \times 0.3 \div 6 \cdot \text{candidate's \#1}$

Confidence for offline UPI is 30%: Play Store SMS rules and UPI 123Pay already existing make success doubtful.

Figure 1.10 • The same ideas, ranked with numbers. Putting even rough numbers on each part flips the candidate's order: the idea they ranked first scores last, because its effort is high and its confidence low. *All inputs illustrative; formula from Intercom's RICE method.*

4 • When it misleads. RICE would tell a family to fix the squeaky door (quick, everyone hears it) before the gas leak in the kitchen (one room, slow to fix). Anything about safety, law or a promise already made goes first and is never scored. RICE also breaks when reach is counted in different units (users a day for one idea, customers a year for another), or when confidence is inflated for a favourite idea.

5 • In the interview. “Let me put rough numbers on reach, impact, confidence and effort, so we can argue about the inputs.” Show one calculation out loud, give the others as results, and say what you do first regardless of the score.

CASE 1.2 • CREDIT FOR TEENAGERS

Product design

Hard IIM B Sigma PM Casebook 2026–27, p. 38

Try it first. Before reading on, answer one question: who legally borrows the money in a “credit for teenagers” product in India?

THE QUESTION

Problem Statement – Create a fintech platform for credit for teenagers.

A I would start by understanding the problem holistically. While teenagers are the primary users, they legally cannot take credit on their own. A guardian must be involved to provide consent and manage credit risks. The platform would hence have two types of accounts- one for teenagers and one for guardians.

Q Sure, that's a fair point. So how would describe your target segment?

A While we will ensure security by giving guardians control over approvals, limits, and spending, the actual target demographic can be split into two groups:

- 13 to 15 years old: Completely dependent on parents, so their credit needs are minimal.
 - 16 to 18 years old: More independent, with daily expenses on transport, food, tuition, and leisure.
- Since 16 to 18-year-olds have higher financial needs, they would be the primary target audience for the platform.

Q This looks good. You may proceed with the features.

A Sure. Teenagers in this age group typically need credit for above-mentioned daily expenses, educational purchases and emergency needs.

For this reason, I would add the following features for teenagers:

1. Pocket-money feature – Guardians can load the account with a fixed monthly allowance.
2. Student transport pass integration – Allows easy purchase of discounted passes for public transport.
3. Gamification for financial responsibility – Rewards good spending habits and credit score improvements.

Q Great. And are there any features you would add for the guardian's persona?

A Yes, for the guardian's account, I would add the ability to set credit limits post which guardian approval would be required and transaction monitoring.

- Q** Can you think of more ways in which you could ensure responsible credit usage and prevent misuse?
- A** By combining parental control with an engaging experience, we can ensure responsible credit usage. I would add the following features to do so:
- Introducing a “Teen Credit Score” to encourage responsible spending.
 - Providing incentives like cashback or discounts for good financial behavior.
 - Granting guardians the ability to block specific merchants and categories.
- Q** And what challenges and trade-offs do you foresee in implementing this product?
- A**
1. Balancing security with user experience – Too many restrictions might frustrate teenagers, while too much flexibility could lead to misuse.
 2. Credit risk for the issuing company – Since teenagers lack financial history, credit default risks are higher.
 3. Adoption hurdles – Convincing both parents and teenagers to adopt the app may require strong trust-building efforts.
- To tackle these, I would focus on a robust onboarding experience, clear communication on security measures, and strong fraud detection mechanisms.
- Q** Fair enough. Now how would you track the success of the platform?
- A**
1. User engagement metrics – Active users, average transactions per month.
 2. Responsible credit usage rates – Improvement in teen credit scores over time.
 3. Guardian involvement metrics – Number of approvals, rejections, and limit changes.
 4. Default rates – Percentage of non-payment cases.
- Q** Thank you, that would be all.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Spotted the legal problem in the first answer:** teenagers can’t take credit alone, so there must be a guardian account.
- ✓ **Split teens by age with a reason** (16–18 spend on their own) and picked one group.
- ✓ **Gave the guardian real controls:** limits, approvals, blocked categories.
- ✓ **Named honest trade-offs** (control vs freedom, default risk, two groups to convince) and put default rate in the metrics.
- ✗ **Never asked who lends, or why.** A bank wanting future customers and a fintech wanting interest income build very different products.
- ✗ **Didn’t follow the law through.** A contract with a minor is void in India (Indian Contract Act, 1872, s. 11), so the teenager can’t be the borrower. The credit has to be the parent’s.
- ✗ **Listed features without the job they do.** A transport pass and pocket money aren’t credit; why does a 17-year-old need to borrow at all?
- ✗ **No North Star.** Four metric families, none marked as the one that decides success.

CHECK · “INTRODUCE A TEEN CREDIT SCORE” **OVERSTATED**

Credit bureaus such as CIBIL score people with loans or cards in their own name, which minors can’t have, so there is no official score to “improve”. An in-app score is still a good idea if it is called what it is: a habit meter that rewards staying within the limit and repaying the parent on time. It becomes useful at 18, when the bank can use that history to offer a first card.

REAL-LIFE EXAMPLE · WHAT INDIAN TEENS CAN USE TODAY

Prepaid teen cards (FamApp, formerly FamPay; Junio) hold money loaded by parents. In June 2022 RBI barred loading such prepaid cards from credit lines, which ended “card-like credit” products built this way. Two newer routes: since July 2025, banks may let children aged 10+ run their own savings accounts within bank-set limits; and **UPI Circle** (2024) lets a parent give a child UPI spending of up to ₹15,000 a month from the parent’s account. Google Pay (“Pocket Money”, Dec 2025) and Paytm built teen features on it.

STRONGER ANSWER



Figure 1.11 · A design answer when two people must both say yes. Moves ② and ③ happen twice, once for each user. *Author’s summary.*

- A** ① **CLARIFY**
- Two questions first. Who is lending, and what do they want from this? And since a minor can’t sign a loan in India, can I assume the parent is the borrower and the teen spends within limits the parent sets?
- Q** It’s a bank. It wants to win customers before they turn 18. Yes, assume the parent borrows.

Why this move: the law and the lender reshape the brief. “Credit for teens” becomes “a parent’s money the teen can use, and a way to learn credit before 18”.

A ② TWO USERS · ③ THEIR JOBS

The 17-year-old's job: "Let me handle my own money without asking every time." The parent's job: "Let my child be independent without me worrying about overspending or scams." The parent can veto everything, so I'll win the parent's trust first (Figure 1.12).

A ③ WHAT HOLDS THEM BACK

Today they use cash pocket money. What pushes them away from it: asking is awkward, and cash gets lost. What pulls them in: paying by UPI like friends. What holds them back: parents fear overspending and scams, teens fear being watched. My features must shrink those fears, more than they add shine (Figure 1.13).

Why this move: in a two-person product, fear is usually a bigger barrier than missing features.

A ④ FEATURES

Allowance on UPI Circle: the parent's account, a ₹3,000 monthly limit, the teen pays by scanning any QR. **"Borrow ahead":** when the allowance runs out, the teen can spend up to ₹500 more, taken from next month's allowance, once the parent approves in one tap. That is practice credit: a limit, a due date, a repayment, and no loan to a minor. **Weekly summary** for both, framed as progress, not surveillance.

A ⑤ RISKS

Scammers target teens with "free game credits" links. So payments to a new payee above ₹1,000 need the parent's approval, and the parent can pause spending with one switch.

A ⑥ MEASURE

Goal (North Star): active parent-teen pairs, meaning the teen paid at least once that week. Illustrative sizing: 20 lakh of the bank's customer families have a 16–18-year-old; 10% sign up in year one = 2 lakh pairs, spending ₹60 Cr a month through the bank. **Safety check (guardrail):** scam losses and the share of teens who hit the limit every month, a sign of money stress. **Trade-off:** more parent control means less teen use; default to alerts, not approvals. **Next:** at 18, offer a first credit card built on the allowance history.

FRAMEWORK · TWO-SIDED USERS AND PERSONAS

1 · The situation. The teen uses the product, but the parent pays, sets limits and can delete the app. Designing for the teen alone produces features the parent will never switch on.

2 · An everyday example. A school canteen has two users: children who eat and parents who pay. Children want samosas; parents want vegetables and a monthly bill with no surprises. Successful canteens offer a prepaid card and let parents block fried food.

Question	School canteen	Teen payments
Who uses it?	The child, at lunch	The teen, at shops and online
Who pays or decides?	The parent, monthly	The parent, who funds and can pause it
What each wants	Tasty food · healthy food, no surprise bills	Freedom · safety and visibility
Who can veto?	The parent	The parent

3 · The framework. List every party, what each one wants, and what flows between them. Then write one **persona** per important party: a short, specific portrait of a real kind of user ("Riya, 17, Class 12 in Pune, spends ₹2,500 a month on food, travel and books; her father checks every SMS alert"). Personas stop you designing for "users" in general.

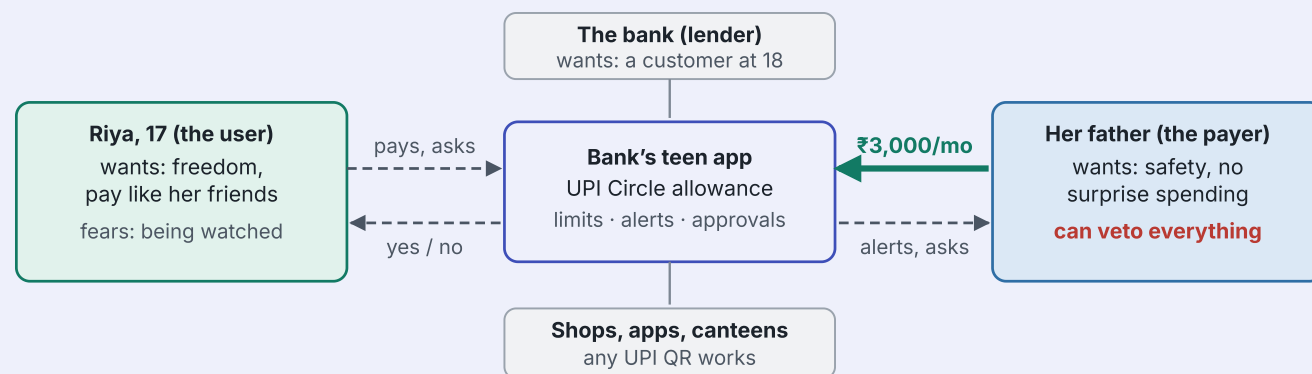


Figure 1.12 · Two users, one product. The teen uses the product; the parent funds it and can switch it off. Every feature has to answer to both, and the parent's "no" ends the product. *Author's model; UPI Circle limits from NPCI (₹15,000 a month maximum).*

4 · When it misleads. Personas become fiction when they are invented at a desk: "Riya loves fintech and budgeting" describes nobody. Base each one on something observed (ten parent interviews, the bank's own data). And two-sided thinking can spiral: five parties with five personas each fills 20 minutes and designs nothing. Two or three parties are usually enough.

5 · In the interview. "There are two users here with different goals, and one of them can veto. Let me take the parent first." Then give each a one-line persona and move on.

FRAMEWORK · JOBS TO BE DONE, AND THE FOUR FORCES

1 · The situation. The candidate listed features (transport pass, gamification). None of them explains why a family would switch from cash pocket money, which works fine today.

2 · An everyday example. Nobody wants a drill; they want a hole in the wall to hang a photo. And even when a better drill exists, people keep the old one: the new one is expensive (anxiety) and the old one is in the cupboard (habit).

3 · The framework. A **job to be done** (JTBD) is the progress a person wants in a situation: “When [situation], I want to [motivation], so I can [outcome].” For the teen: “When I’m out with friends, I want to pay my share myself, so I don’t have to call Dad.” Whether they switch depends on **four forces**: two pushing towards the new product and two holding them back. Switching happens only when push plus pull beats anxiety plus habit.

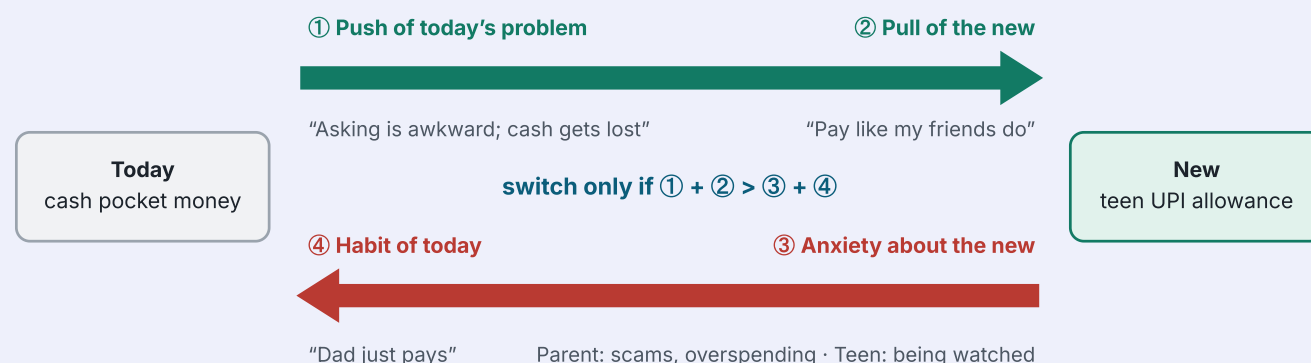


Figure 1.13 · The four forces on a family deciding whether to try a teen payments app. Green forces pull towards the new product, red ones hold the family back. Most teams add pull (rewards, design); the cheaper win is often cutting anxiety (limits, approvals, a pause switch). *Four forces after Bob Moesta and Chris Spiek (JTBD “forces of progress”); family examples illustrative.*

Force	Buying a new drill	Teen payments
Push	The old drill can't go through concrete	Awkward asks, lost cash
Pull	The new one is cordless	Paying like friends, feeling grown-up
Anxiety	"Will I use it enough for ₹6,000?"	Scams, overspending, surveillance
Habit	The old drill is right there	"Dad just pays"

4 · When it misleads. Jobs written too broadly become useless: “help teens manage money” fits every fintech app ever built. Keep the situation specific (“out with friends, bill arrives”). And the four forces explain switching, not staying: a family that tried the app and left after a month has a retention problem, which needs its own diagnosis.

5 · In the interview. “The teen’s job is to pay their share without calling a parent; the parent’s job is to allow that without worry. The parent’s anxiety is the biggest force against us, so my first features reduce it.”

CASE 1.3 · IMPROVE YOUR FAVOURITE UPI APP

Product improvement

Medium IIM B Sigma PM Casebook 2026–27, pp. 116–117

Try it first. Pick the app you use most. In 5 minutes, name one group of users who might leave it, and the single moment that would make them leave.

THE QUESTION

What is your favourite UPI application and how would you improve it? (AppDynamics)

A I'll start with introducing PhonePe so that we are on the same page. PhonePe is a payments app that meets all payment, investment, mutual funds, insurance and banking needs. They have moved from being a simple payments app to being a complete financial services platform, recently adding wealth management solutions to their portfolio. It has a super app platform called Switch with integration across multiple merchants related to food, travel, lifestyle, etc. Another interesting product is an interactive website called PhonePe Pulse showing data, insights, and trends on digital payments done through its app. Their major competitors would be GPay, PayTM, and Cred.

Q What would be the main differentiators for PhonePe?

A PhonePe has not restricted themselves to only UPI and are making financial services products accessible to people, be it insurance, mutual funds, or gold. For investing needs, they have interesting features like SIP and wealth calculator, Super Funds and multiple other educative resources to help investors. The UI is very intuitive and clean. The buttons and icons are also well-spaced out with a big font, making them readable and accessible even for elderly. In terms of the home page, I noticed a difference between GPay and PhonePe. For GPay, the focus seems to be on sending and receiving money and the frequent people we've transacted with being on the top of the home page. On PhonePe, the home page has multiple sections - Recharge, Insurance, Switch, Investment. Also, compared to GPay, PhonePe has a higher payment success rate. Can I now proceed with analyzing how they can improve their product?

- Q** Yes, please proceed.
- A** I would like to know what is the goal of the app improvement? Improving user engagement, increasing market share, increasing user retention or increase revenue?
- Q** What is the most pressing issue in the payments space according to you?
- A** The digital payments market is mature but has less customer loyalty and stickiness. Considering this, I would like to consider user retention as the goal of improvement, which would ultimately translate to an increase in revenue.
- Q** That sounds fair. Go ahead.
- A** I'll now move on to identifying the user personas for the product. The personas I can think of are:
1. Students
 - Age group 15-21, tech savvy, usually transact using online payment channels, use the app to send money or split expenses among friends.
 - Saving money is their primary objective and they are attracted to cashbacks and discounts.
 - Pain point: They are most affected when they face payment failures if bank servers are down, since they don't usually have cash on hand and are in a hurry to complete transactions. They don't find the current offers on PhonePe lucrative.
 2. Young Curious Investors
 - Age group 21-25 years, have disposable income which they are looking to invest, are new to the investment world and would like to know how and where to invest.
 - Pain Point: They don't know which funds to invest in. Also, trust on making investments using PhonePe is low.
 3. Working Professionals
 - Age group 25-40, comparatively loyal to the app, use the app to pay bills, for recharges, buying insurance, booking travel tickets, etc.
 - Pain Point: Amidst their schedule, they forget due dates for utility bills. They have to use multiple apps for recharge, bill payment and booking needs.
 4. Parents/Grandparents
 - Aged 55+, not comfortable with using mobile apps, use the app to send and receive money.
 - Pain Point: They tend to forget the sequence of steps to make a transaction. Also, trust on online payments is low.
- Do these personas look fine to you or do you want me to consider anything else?
- Q** This seems good, you can go ahead and prioritize a persona.
- A** Sure. Though the parents' segment has a differentiated need and might present as a lucrative segment, they have trust issues. This requires a different approach and the app features wouldn't be able to contribute much to this. I would not prioritize this segment for now. Based on my understanding of the needs of the other users, I would prioritize students and young working professionals, as there is no technological barrier for them. Also, 66% of the Indians fall under this bracket and forms the majority of the market, hence prioritizing makes more sense. Are you fine with me proceeding with this or would you like me to prioritize a different persona?
- Q** The target personas seem fine. How would you solve for the problems?
- A** I will now suggest 2 features for these 2 target segments, plus an additional delight feature that can cater to all our customers and act as a differentiator.
1. RewardPoints/Cashback: Assured points after a certain transaction amount and vouchers across more categories (like Cred). The points can be redeemed at a certain conversion rate. However, this might burn some capital and collaborations with brands need to be sorted out.
 2. Reminders & Auto-Payments: Payment reminder notifications can be sent to users. If somehow the users ignore this, then payment mandates to top-up PhonePe wallet and make the payments on behalf of the user can be done to ensure that the credit scores are not affected.
 3. (Delight) Fail-proof transactions: Sometimes, UPI transactions fail in crunch moments. Many users end up paying double the amount and have to haggle for refunds later. This is particularly problematic in situations where the receiver is a person with whom we won't be transacting again (like public transport/roadside stalls). For such situations, PhonePe can create a suspense wallet in the backend, which will act as a middle layer between the sender and the receiver. When a transaction is initiated, the amount is credited to this wallet first and then sent to the receiver's bank. In case the transaction fails, the suspense wallet would retry the payment after some time, however, the receiver would receive a notification stating that the transaction has been made. This is similar to durable execution seen in workflow engines like Temporal.
- Q** This seems innovative, but difficult to implement.

A The efforts would be high initially, but once implemented, this can result in nearly 100% transaction success rate. We can also gatekeep this technology and keep it exclusive for PhonePe users. Later on, as the competition catches up, the ecosystem becomes better and it's a win-win for all customers.

Q Fair, what all metrics do you plan to track?

A I would look at the user retention rate, the daily and monthly active users and the number of transactions on an average per customer.

For the delight feature, I would look at transaction success rate and number of complaints related to payment failures.

Q That sounds alright. Thanks, we are done with the case.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Described the product and its rivals first**, so both sides were talking about the same PhonePe.
- ✓ **Asked for the goal**, and when the interviewer turned it back, chose retention with a reason: payments have little loyalty.
- ✓ **Gave each persona a pain point** and dropped one group openly, with a reason.
- ✓ **Aimed a bold idea at the biggest pain** (failed payments at a counter) and gave it its own metrics.
- ✗ **Two numbers don't hold**: the 66% share and "nearly 100% success". And "PhonePe has a higher success rate than GPay" can't be checked: NPCI publishes success rates by bank, not by app.
- ✗ **Never ranked the three ideas**. Cashback is the most expensive way to buy loyalty, and rivals copy it within a week.
- ✗ **Defined retention loosely**. Retained over which window: next week, next month? Measured on payers or app openers?
- ✗ **No safety check**. A "fail-proof" payment that tells the receiver "paid" early invites fraud; nothing in the metrics would catch it.

CHECK · "66% OF INDIANS FALL UNDER THIS BRACKET" (STUDENTS AND YOUNG PROFESSIONALS) **OVERSTATED**

About 68% of Indians are aged 15–64, the whole working-age band (UN World Population Prospects 2024). The candidate's groups, roughly 15–40, are closer to 4 in 10 Indians. Better still, size the market you care about: UPI users, not the population.

CHECK · "A SUSPENSE WALLET GIVES NEARLY 100% TRANSACTION SUCCESS" **DOESN'T HOLD UP**

Three problems. **(1)** Most failures happen at the payer's bank (the issuer): if Asha's bank is down, her money can't leave her account, and a middle wallet doesn't change that. **(2)** Telling the shop "paid" before money arrives means PhonePe is lending against a pending payment, and fake "paid" screens are already a common scam. **(3)** Holding customers' money in a wallet needs an RBI licence as a payment aggregator or prepaid-instrument issuer. The pain behind the idea is real, and the Stronger answer keeps it.

STRONGER ANSWER

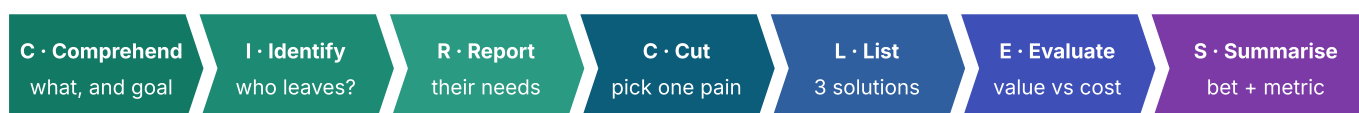


Figure 1.14 · CIRCLES as seven moves. An improvement answer that walks these in order rarely gets lost. Spend the least time on the first three and the most on Evaluate. *CIRCLES method: Lewis C. Lin, Decode and Conquer (2013).*

A C · COMPREHEND

PhonePe starts 45.9% of UPI payments, and earns from what those payers buy next: loans, insurance, bills. Switching is free, because any UPI app scans any QR. So I'd pick one goal: keep monthly payers paying next month. Is that fair?

Q Retention is fine. Go ahead.

A I · IDENTIFY · R · REPORT

Who leaves? Probably students and first-jobbers: they chase cashback, carry no cash, and pay small amounts many times a day. Their need: "When I pay at a counter, it must go through, or I must be able to show it will."

Why this move: one sharp need beats four broad personas. It tells you what to build.

A C · CUT

Of their pains (failed payments, weak offers, splitting bills), failures hurt most. One failed payment in front of a shopkeeper is remembered longer than ten good ones.

A L · LIST

Three ideas. **Smart switch**: if the user's bank is failing right now, suggest their second linked bank or UPI Lite before they tap Pay. **Pending proof**: a clear pending screen with "money back by tomorrow if not credited", and the shop's soundbox says "payment pending", not silence. **Split and remind**: split a bill and auto-remind friends.

A E · EVALUATE

A successful payment is a basic: nobody thanks you when it works, and everyone leaves when it fails (Figure 1.15). Smart switch protects that basic and is cheap, since PhonePe already tracks each bank's health. Size it, illustratively: if 1% of PhonePe's 11.25 bn monthly payments fail at the user's bank, and a third are saved, that's 3.75 crore payments a month. Pending proof comes next; splitting is a nice-to-have.

Why this move: ranking by "basic vs nice-to-have" plus a rough size is what turns a list into a decision.

A S · SUMMARISE

Bet: build Smart switch first. **Goal (North Star):** share of this month's payers who pay again next month. **Safety check (guardrail):** double debits and complaints about unwanted prompts. **Trade-off:** prompting a second bank can feel like nagging, so show it only when the user's bank is failing. **Next:** test on 5% of users against a control group.

FRAMEWORK · CIRCLES: A ROUTE MAP FOR "IMPROVE X"

1 · The situation. "How would you improve your favourite app?" is so open that candidates either ramble or jump to a feature in the first minute. The interviewer wants to see how you get from a product to one well-chosen bet.

2 · An everyday example. Your hostel asks you to improve the college canteen. You'd first ask what it is for (cheap, fast lunches), who eats there (hostel students, day scholars, staff), what annoys them, which annoyance matters most, some fixes, which fix is worth it, and then you'd propose one.

Move	College canteen	PhonePe
Comprehend	Cheap, fast lunch between classes	Payments habit that sells loans and bills; goal: retention
Identify	Hostel students, day scholars, staff	Students and first-jobbers most likely to leave
Report	"The queue eats my lunch break"	"It must go through at the counter"
Cut	Queue time matters most	Failed payments matter most
List	Pre-order app, second counter, combo meals	Smart switch, pending proof, split and remind
Evaluate	Second counter: cheap, halves the queue	Smart switch: cheap, protects the basic
Summarise	Open a second counter at 1 pm; measure wait time	Ship Smart switch; measure month-on-month payer retention

3 · The framework. CIRCLES is seven moves (Figure 1.14): Comprehend the product and goal, Identify the customer, Report their needs, Cut to one, List solutions, Evaluate trade-offs, Summarise. Its value is the order: understanding before ideas, and a decision at the end.

4 · When it misleads. Recited mechanically, CIRCLES eats the clock: a candidate describes the product, lists four personas with ages, prioritises two, and the interviewer says "we have five minutes left" before the first idea. Spend about two minutes on C-I-R together, and most of your time on L and E.

5 · In the interview. Don't name it. Say: "I'll pick a goal and a user, find their biggest pain, then compare a few fixes." The structure shows without the acronym.

FRAMEWORK · KANO: BASICS, PERFORMANCE, DELIGHTERS

1 · The situation. The candidate offered cashback, reminders and a fail-proof payment as if they were the same kind of thing. They aren't: one fixes anger, one adds value step by step, one surprises.

2 · An everyday example. A hotel room. Hot water is a **basic**: no one praises it, everyone complains without it. Wi-Fi speed is **performance**: faster is steadily better. A handwritten welcome note is a **delighter**: nobody expects it, and it makes the stay memorable. Next year, if every hotel leaves a note, it stops delighting.



Basics: expected; missing = anger

1 Payment goes through · 2 never debited twice
3 "Paid ✓" within seconds

Performance: better = happier

4 Fewer taps to pay · 5 clear pending status
6 Bill reminders before due dates

Delighters: unexpected wins

7 UPI Circle for family · 8 "Pending proof"
9 Scratch cards: delight in 2017, now expected; cutting them feels like a loss (dashed arrow)

Rule: fix basics before adding delight.

No reward outlives a failed payment.

Figure 1.15 · Kano's three curves, with real UPI features on them. Basics (red) can only prevent anger, performance features (blue) add satisfaction step by step, delighters (green) surprise. Over time delighters slide down into basics, as scratch cards have. Kano model: Noriaki Kano et al. (1984). Placement of features is the author's judgement.

3 · The framework. Sort each feature by how satisfaction responds when you do it better. Fix basics first, compete on performance, and use delighters sparingly, because they fade into basics as rivals copy them.

4 · When it misleads. The same feature sits in different places for different users: UPI Lite (no PIN for small payments) delights a student buying chai and worries a 60-year-old who wants a PIN for everything. And the labels are guesses unless you ask users; Kano's original method is a short survey pairing "how would you feel if it had this?" with "if it didn't?".

5 · In the interview. "Payment success is a basic here. Nobody will stay for rewards if payments fail, so I'd protect the basic first and treat cashback as a delighter we can't afford to depend on."

CASE 1.4 · FRAUD PREVENTION IN A BANKING APP Product improvement Medium IIM B *Sigma PM Casebook 2026–27*, pp. 110–111

Try it first. Before reading: in India, where do people lose the most money to payment fraud, and how are they usually tricked? Then sketch one fix.

THE QUESTION

Can you improve the IDFC Bank mobile app for preventing frauds

- A** Sure, first, I would like to understand if we are looking at any specific type of fraud and bank. Moreover, are we looking at any specific region or platform that needs improvement.
- Q** You can assume the app to be a standard mobile banking app like IDFC bank. We are looking at frauds during any transactions. There is no constraint on the region since the app is used pan – India.
- A** Sure. So I will proceed in my analysis with the IDFC bank mobile app, keeping in mind the Indian context. Are there any metrics that we are targeting to improve?
- Q** You may proceed. What metrics should we target according to you.
- A** Since we are targeting fraud prevention, I believe the metrics should be focusing on improvement in the fraud cases while not losing any customers due to the added feature. It would be even better if we could get more customers. We should target customer retention and engagement. The L1 metrics here, would be DAU and number of frauds prevented. I will come up with L2 metrics after the solution. Do we have any user personas in mind or should I come up with them?
- Q** Sounds good. You can come up with user personas.
- A** I would like to divide the user persona into two parts Merchants and Regular Users, further divide them into different categories.
- A Merchants**
1. General Stores: Large volume of transactions with low average order value.
 2. Large Stores: Low volume of transactions with large average order value.
- A General Users**
1. Tech Savvy: Aware of the financial processes and frauds that occur commonly. Includes working professionals.
 2. Non – Tech Savvy: Unaware about how to navigate the financial environment. Includes non-technical people like the elderly.
- I would like to prioritize the general stores due to the large number of transactions and elderly as these segments are more prone to frauds.
- Q** The user personas are well put but I would like you to focus on the Tech Savvy Users.
- A** Alright. As a Tech Savvy User myself, I can think of several pain points that might come up here .
1. There are times when I receive an OTP for a delivery, receiving food or login from a friend. I blindly forward the OTP on call if I am occupied in a meeting.
 2. I find the tap to pay feature on credit cards particularly risky and prone to unwanted transactions.
 3. Finally, I also believe an error could be made by transferring funds to an incorrect phone number through UPI
- Q** Can you elaborate more about the Tap to pay problem
- A** I find the tap to pay feature in credit cards to be a weak link in the transaction process. Several things can go wrong here, and the most common one would be an unintended transaction. Let's say I have my card in my back pocket wallet. A fraudster can just bring the card reader close to the wallet and initiate a transaction
- Q** That is a fair problem but how would this relate to the mobile banking app.

A This can be solved by a notification based approach. Whenever a tap to pay transaction is initiated, there would be a notification on the mobile banking app to accept or reject the transaction. The notification should include transaction details like location and recipient. The user can then choose to accept or reject the transaction from the notification. However, since the tap to pay model is designed for convenience, the transaction should go through successfully if there is no response to the notification for 30 min.

Q Tap to pay is indeed a feature for convenience. This would impede users who want quick transactions. How would you tackle this? Also describe the flow.

A We can have a toggle for this feature. Users who don't want this feature can disable it completely, but I believe a significant number of user should benefit from this. The flow would be a notification as a banner with limited details with an accept or reject button. If a user taps on the notification, he is redirected to the mobile banking app and all details of the transaction and recipient are shown.

Q That is a satisfactory solution. Is there anything else you would like to add?

A Yes, to track if this feature is successful at preventing frauds, we can track a few metrics. The first one would be the number of users who have this feature enabled. The second metric would be the number of times a 'reject' option is chosen in the notification. This metric would directly tell us the number of fraudulent or unintended transactions that were performed. I would also like to add that here, we tackle the case of someone else using a credit card with the owner's consent. Let's say a child is using the fathers credit card for a tap to pay transaction, it gives a chance for the father to check if the transaction is ok to go ahead with. That would be all.

Q Thank you

WHAT WORKED · WHAT WAS MISSING

- ✓ **Clarified fraud type, platform and region** before starting.
- ✓ **Set two goals at once:** fewer frauds, and no customers lost to the added step. That second goal is a guardrail in all but name.
- ✓ **Followed the interviewer's steer** to tech-savvy users without arguing.
- ✓ **Designed a concrete flow** with an opt-out toggle, and noticed a child using a parent's card.
- ✗ **Picked a small slice.** In FY25 India saw 12.64 lakh UPI fraud cases against about 66,000 credit-card fraud cases. The candidate's own first pain, sharing an OTP on a call, is where most money goes.
- ✗ **The feature can't work as designed:** card payments are approved in seconds, not 30 minutes.
- ✗ **"Rejects = frauds" overcounts.** Rejects include mistaken taps and family use; nothing counted genuine users annoyed.
- ✗ **Missed what exists.** Since January 2020 RBI has required banks to let customers switch contactless, online and international use on or off, and set limits, in the app.

CHECK · "APPROVE TAP-TO-PAY FROM A NOTIFICATION; AUTO-APPROVE AFTER 30 MINUTES" **DOESN'T HOLD UP**

A card payment is approved or declined at the terminal in a few seconds. A shop won't hand over goods on a 30-minute maybe, and a fraudster's tap would be approved long before the notification arrived. The skimming worry is also overstated: taps without a PIN are capped at ₹5,000, and the thief needs a merchant account that can be traced. What works: contactless off or a low tap limit set in the app, plus an instant alert with a one-tap "block card and report".

STRONGER ANSWER



Figure 1.16 · A fraud answer in six moves. Move ⑤ is the one most candidates skip, and the one a bank's leadership asks about first. *Author's summary.*

A ① **CLARIFY**

Which fraud: cards, UPI or net banking? And whose loss is it: do we refund only when an account is hacked, or also when a customer is tricked into paying?

Q Your call. The bank refunds only unauthorised fraud, but tricked customers blame us anyway.

A ② **SIZE**

Then I'd go where the losses are. In FY25 India had 12.64 lakh UPI fraud cases worth ₹981 Cr, against about 1.45 lakh card and net-banking cases worth ₹822 Cr. Most UPI fraud is people being talked into paying: fake customer care, "refund pending, enter your PIN". So I'll focus on tricked payments.

| Why this move: one sentence of sizing moves the whole answer to where the money is lost.

A ③ **MOMENT**

Tech-savvy users get tricked too, usually while on a call with the scammer or sharing their screen. The moment to step in is after they type the amount and before they enter the PIN.

A ④ DESIGN

Friction only when risk is high. **On a call or screen-sharing, paying a new payee:** a full-screen warning in plain words and a 10-second pause. **First payment above ₹10,000 to a new payee:** a one-hour cooling-off the user can cancel, similar to what RBI proposed in April 2026. **A kill switch:** one tap freezes all outgoing payments.

A ⑤ SAFETY CHECK

Illustrative numbers: 1 crore users make 3 crore transfers a month. If our rules flag 1%, that's 3 lakh. If 5% of flagged payments are fraud, we stop 15,000 frauds and delay 2.85 lakh genuine payments: 19 good payments slowed for every fraud stopped. That ratio is my guardrail (Figure 1.17).

Why this move: every fraud rule also hits honest people. Naming the cost, with a number, is what makes the design credible.

A ⑥ MEASURE

Goal (North Star): ₹ lost to fraud per ₹1 lakh moved, down 30% in two quarters. **Safety check (guardrail):** genuine payments delayed per fraud stopped, and the share of warned users who give up a payment that was real. **Trade-off:** stricter rules catch a little more fraud and slow a lot more honest payments. **Next:** start with 5% of users and tune the rule weekly.

FRAMEWORK · GUARDRAILS AND COUNTER-METRICS

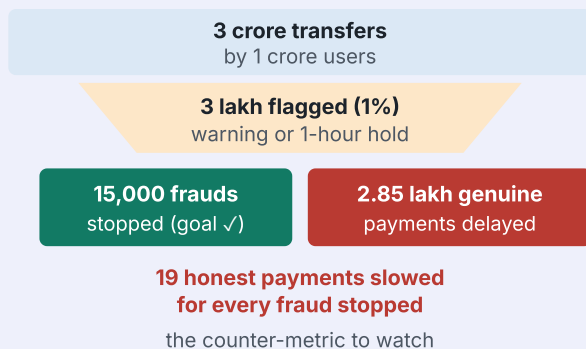
1 · The situation. Any fraud rule can drive fraud to zero: block every payment. The real question is what the rule costs honest users, and whether anyone is watching that cost.

2 · An everyday example. A school wants fewer late-comers, so the gate is locked at 8:00 sharp. Late arrivals drop to zero within a week. But children who would have arrived at 8:05 now stay home, and attendance falls. The school fixed the number it watched and damaged the one it didn't.

Part	School gate	Fraud rules
Goal metric	Late arrivals ↓	₹ lost to fraud ↓
Counter-metric	Attendance (children sent home)	Genuine payments delayed or abandoned
The dial	Gate time: 8:00, 8:10, 8:15	Share of payments flagged: 0.5%, 1%, 3%
Right setting	Most children in class, on time	Least total harm: fraud plus friction

3 · The framework. A **guardrail** is a metric you promise not to hurt while chasing your goal, with a limit written down in advance ("genuine payments delayed stay under 1%"). A **counter-metric** is the specific number that would move if your change backfired. Pick them before launch, put them on the same dashboard as the goal, and agree what happens if they cross the line.

One month, 1% of payments flagged (illustrative)



Turning the dial: how strict?

flagged	frauds caught	genuine delayed
0.5%	40%	1.4 lakh
1%	71%	2.85 lakh
3%	90%	8.8L

Going from 1% to 3% catches 3,900 more frauds and slows 6 lakh more honest payments.
Without the counter-metric, "90% caught" looks like a win.

Figure 1.17 · The goal and its counter-metric move together. Left: what one setting does in a month. Right: stricter settings buy small gains in fraud caught at a large cost to honest users. The guardrail decides where to stop. *Illustrative model: 3 crore transfers a month, ~21,000 of them fraudulent.*

4 · When it misleads. An airport adds a second bag check. Smuggling falls, queues reach three hours, and passengers miss flights. If "missed flights" isn't on the dashboard, the change looks like a success. Guardrails fail the same way when they are set but never read, or when there are so many (twelve "must not drop" metrics) that nothing can ever ship. Two or three, with written limits, is plenty.

5 · In the interview. "My goal is fewer rupees lost to fraud. To make sure I'm not buying that with honest customers' time, I'll watch genuine payments delayed per fraud stopped, and loosen the rule if it goes above about 20."

CASE 1.5 · PROMOTIONS IN GOOGLE PAY

Metrics Medium IIM A Product Mindset 2026–27, p. 294

Try it first. Before any metric, answer: why does Google want partner coupons inside Google Pay at all? Then name the one number you would report to its head.

THE QUESTION

You are a product manager in the Google Pay team. You have been given the task of defining the metrics for tie-in promotions on Google Pay. How will you go about it?

A Before I begin, let me confirm my understanding of the feature. Google Pay rewards users with scratch cards after eligible payments. These rewards may be cashback or coupons from partner brands like Lenskart, Mamaearth, Zomato, etc. These partner coupons are tie-in promotions. Is that accurate?

Q Your description of tie-in promotions is accurate. Essentially tie-in promotions involve a company partnership with another company for promotions. You can proceed further with your analysis.

A Great. The metrics should align with the business goals.

- Users should find the promotions useful enough to continue using Google Pay.
- Promotion partners should benefit through customer acquisition and revenue.

Accordingly, I'd define metrics from the perspective of users and promotion partners.

Q The business goals seem accurate. Please proceed with the metrics.

A *The casebook prints the previous two turns a second time here; the repeat is removed.*

User Metrics : The user metrics should measure the frequency, utilization and effectiveness of promotions.

Success Formula : No. of successful uses of promotions = No. of payments × % of tie-in promotion scratch cards × % of tie-in promotions used

Key metrics:

- A**
- Number of payments / payment frequency
 - % of scratch cards that are tie-in promotions
 - % of promotions that expire unused
 - Average number of brands/promotions shown per user
 - % of repeat promotions from the same brand/type

Promotion Partner Metrics : From a partner's perspective, success depends on how effectively Google Pay drives customer engagement.

Success Formula : No. of successful uses of a specific promotion = No. of payments × % of promotional coupons for a given partner × % of that specific coupon × % of users successfully using the promotional coupon

Key metrics:

- Number of tie-in promotion partners
- Partner retention rate
- Partner churn rate
- Average coupons shown per partner
- % of coupons expiring unused (overall and partner-wise)
- Average coupon recipients per campaign
- % of users redeeming a partner's promotional coupon

These metrics help identify which promotions are performing well and which need to be redesigned or discontinued.

Q I think this covers most of the important metrics for tie-in promotions.

WHAT WORKED · WHAT WAS MISSING

- | | |
|--|---|
| <ul style="list-style-type: none">✓ Confirmed the feature in plain words before measuring it: scratch cards, cashback, partner coupons. A model “confirm the thing”.✓ Named both sides: users and partner brands.✓ Wrote the metric as a formula, which is a driver tree in words.✓ Gave partners their own formula and metrics, including churn. | <ul style="list-style-type: none">✗ Never asked why Google runs the promotions: more payments, partner revenue, or both? Without that there is no North Star, just 15 metrics of equal weight.✗ Never asked whether the coupons cause anything. Would the user have bought from Lenskart anyway? Only a holdout group answers that.✗ No cost or safety checks: coupon farming (₹1 payments to friends to win cards), “not interested” taps, trust.✗ The formula starts one level too low: “number of payments” is really payers × payments per payer, two levers owned by different teams. |
|--|---|

CHECK · “SUCCESSFUL USES = PAYMENTS × % TIE-IN SCRATCH CARDS × % USED” **DIRECTIONALLY RIGHT**

It is a proper driver tree: each factor can be owned and moved by a team. Two caveats. The factors aren't independent (people who pay more also redeem more coupons), so improving one can shift another. And the output is a count of redemptions, which says nothing about whether each redemption was extra business for the partner.



Figure 1.18 · A metrics answer in six moves. Metrics cases reward depth on a few numbers over a long list. *Author's summary.*

A ① WHY?

Why does Google Pay carry partner coupons? My guess: partners fund the rewards, so Google gets rewards that keep people paying on Google Pay without paying for them itself, and partners get customers. Is the priority more payments on Google Pay, or partner revenue?

Q Both, but more payments on Google Pay comes first.

Why this move: the answer decides the North Star. Without it, every metric looks equally important.

A ② WHO GAINS · ③ NORTH STAR

Users want to be rewarded for paying here. Partners want new customers cheaper than ads. Google wants Google Pay to be the default way to pay. So my **Goal (North Star)** is monthly payers who make 10 or more payments, a habit measure. Promotions are one input to it, alongside payment success and new users.

A ④ DRIVER TREE

Illustrative numbers: 10 crore monthly payers × 25 payments = 250 crore payments. 20% earn a scratch card = 50 crore cards. 30% of cards are partner coupons = 15 crore coupons. 2% are redeemed = 30 lakh redemptions a month. The weak link is redemption: 98% of coupons go unused, so I'd fix relevance before signing more partners (Figure 1.19).

Why this move: multiplying the chain out shows which factor is weakest, and that factor becomes the team's focus.

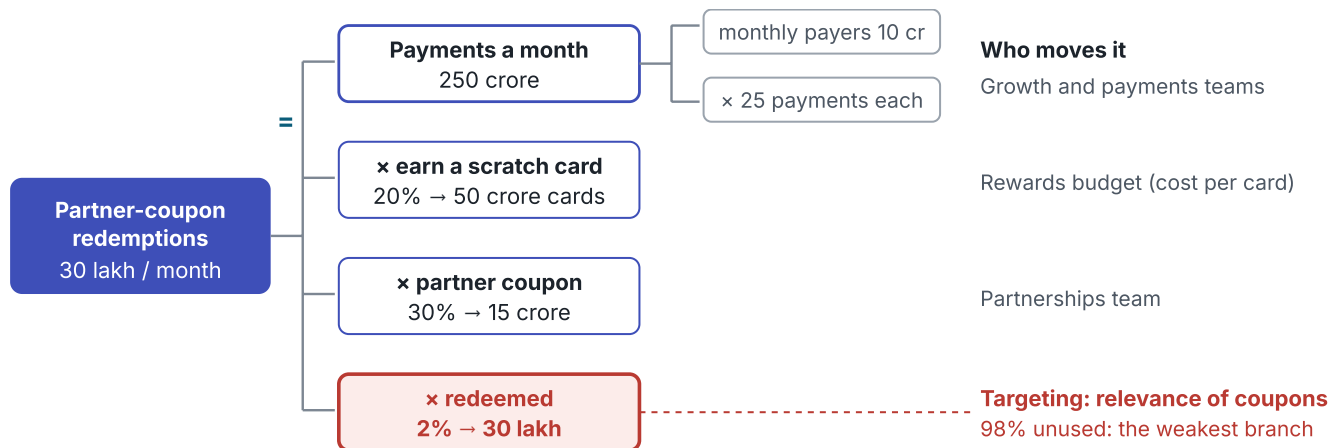


Figure 1.19 · The promotions driver tree, with numbers. Read it left to right: each box multiplies the one above. The red branch is the weakest link, so doubling redemption from 2% to 4% adds as much as doubling every partner deal. *All numbers illustrative. Structure from the candidate's formula (IIM A p. 294), extended one level.*

A ⑤ PROVE CAUSE

Do coupons cause anything? Keep 5% of users on no partner coupons for a quarter. If they make 24.5 payments a month and everyone else makes 25, coupons add 2% more payments. For partners: if 1.2% of users shown a Lenskart coupon buy, against 1.0% of similar users not shown one, only 1 in 6 redemptions is new business. That is what Lenskart should pay for (Figure 1.20).

A ⑥ GUARDRAILS

Safety check (guardrail): coupon farming (very small payments between the same few people), “not interested” taps, and support complaints. **Trade-off:** better targeting means fewer coupons for some users, which can feel like a cut in rewards. **Next:** report incremental purchases per partner every month, not raw redemptions.

FRAMEWORK · NORTH STAR, INPUT METRICS AND THE DRIVER TREE

1 · The situation. The candidate listed 15 metrics. A team can't steer by 15 numbers; it needs one that captures the value it creates, and a map of the levers that move it.

2 · An everyday example. A cricket team's goal is matches won. The captain can't order “win more”, but can work on the things that add up to it: runs per over, wickets, catches. And runs themselves split further: balls faced × strike rate.

Part	Cricket team	Google Pay promotions
North Star	Matches won	Monthly payers making 10+ payments
Input metrics	Runs per over, wickets, catches	Payments per payer, coupon redemptions, payment success
Driver tree	Runs = balls faced × strike rate	Redemptions = payments × card rate × partner share × redemption rate
Weak link	Dropped catches	98% of coupons unused

3 • The framework. A **North Star metric** is the one number that best captures the value users get, and that predicts the business's future. **Input metrics** are the few numbers teams can move directly that feed it. A **driver tree** writes the link as arithmetic, a multiplication or a sum, so you can see which branch is weakest and who owns it (Figure 1.19).

4 • When it misleads. Choose a North Star that can be gamed and it will be. If it were “number of payments”, users would split one ₹500 payment into five of ₹100 to win more scratch cards, and the number would rise while nothing real improved. Driver trees mislead when the branches affect each other: showing more coupons (partner share ↑) can lower the redemption rate, because people ignore a cluttered rewards page.

5 • In the interview. “I’ll pick one number that shows users are getting value, then break it into the three or four levers that drive it, and find the weakest one.” Say the North Star in a sentence before listing any other metric.

FRAMEWORK • HOLDOUTS AND INCREMENTALITY: DID WE CAUSE IT?

1 • The situation. Lenskart sees 30,000 coupon redemptions and is delighted. But many of those buyers were already planning to buy glasses. Lenskart should pay only for the purchases the coupon *caused*.

2 • An everyday example. A coaching centre advertises that its students average 90% in the board exam. But the students who join coaching are often the most motivated ones anyway. The fair test: compare them with similar students who didn’t join.

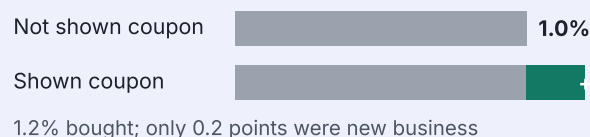
Part	Coaching centre	Partner coupon
Treatment group	Students who joined	Users shown the Lenskart coupon
Holdout (control)	Similar students who didn’t join	Similar users, picked at random, not shown it
Raw result	90% average	1.2% of users bought
Incremental effect	90% – 84% = 6 points	1.2% – 1.0% = 0.2 points: 1 in 6 buyers

3 • The framework. A **holdout** is a random slice of users kept away from a feature so they show what would have happened without it. **Incrementality** is the difference between the treated group and the holdout: the effect the feature caused. Because the slice is random, the two groups are alike in everything except the feature.

For Google: payments per user a month



For Lenskart: share of users who bought



30 lakh redemptions ≠ 30 lakh new sales.

If 1 in 6 redemptions is incremental, the partner’s true result is about 5 lakh extra purchases a month, and that is the number to price the partnership on.

Figure 1.20 • Raw results vs the part a feature caused. Both panels ask the same question: compared with a random group that didn’t get the coupon, what changed? Most of the bar would have happened anyway. *Illustrative numbers.*

4 • When it misleads. Holdouts cost something: those users miss rewards, and the business gives up whatever the feature earns on them. Too small a holdout gives noise: 5% of 10 crore users is 50 lakh people and plenty; 5% of a 2,000-user pilot is 100 people and tells you nothing. And holdouts run for too short a time miss slow effects, such as users getting tired of coupons after two months.

5 • In the interview. “Redemptions tell us coupons were used, not that they caused anything. I’d keep a random 5% on no partner coupons and compare purchases and payments against them.”

CASE 1.6 • ANDROID DAILY USERS DROP 40%

Root cause analysis

Medium

IIM A Product Mindset 2026–27, p. 308

Try it first. Read only the question in the blue box. Take 5 minutes and say your structure out loud before you read the candidate’s answer.

THE QUESTION

You’re the PM for our mobile banking app. Android Daily Active Users (DAU) fell 40% in the last two weeks while iOS DAU stayed steady. Walk me through how you’d diagnose the root cause.

A Let me confirm the issue: Android DAU down 40% over two weeks, iOS flat. Correct?

Q Yes, that’s correct.

A A few clarifiers. Is DAU unique users who opened the app at least once that day? Is the drop across both new and existing users? And is it global or localised?

- Q** Yes, unique daily users. Both new and existing are hit. It's global but more pronounced in North America.
- A** First, metric reliability - any change to analytics tooling or event definitions, or any processing delay or mis-aggregation?
- Q** No changes, and no delays or aggregation errors.
- A** Given accurate metrics, external factors: any industry shift hitting Android specifically, or competitor promotions targeting Android users?
- Q** No significant external events or competitor promotions.
- A** Ruling those out, let me segment deeper. Is it consistent across all Android versions, or specific to one? And across all manufacturers, or specific ones?
- Q** It's primarily Android 12 and above, regardless of manufacturer.
- A** Now internal changes - sudden drops usually coincide with a release. Did we ship a new Android version around when this started, with UX, backend, security or permissions changes?
- Q** Yes - a major Android update two weeks ago, specifically for security and permissions. No equivalent iOS update.
- A** That aligns with the timeline. Release quality next: did crash reports or complaints spike post-release, and did the Play Store rating drop?
- Q** Yes - crash reports surged, specifically at app startup, and negative ratings mentioning crashes rose sharply.
- A** Then the crash logs: a consistent error across devices, correlated with the new permissions logic?
- Q** The logs show repeated crashes tied to permission checks during launch, specifically on Android 12+.
- A** Before I conclude, one loose end. If the bug is purely Android-12+ permission logic, the drop should be roughly uniform wherever Android 12+ is common - yet it's more pronounced in North America. The most likely reconciliation is a staged rollout (the update reaching North America first) or higher Android-12+ penetration there. Can we confirm the rollout sequence and the regional OS-version mix?
- Q** Yes - the update was rolled out to North America first.
- A** To summarise the path: metrics verified; external factors and competitors ruled out; segment isolated to Android 12+ globally, surfacing first in North America via a staged rollout; cause narrowed to our recent security/permissions update crashing the app at startup on Android 12+.

Solutions:

Short term: roll back to the last stable version; if rollback isn't feasible, ship a hot-fix correcting the permissions logic and deploy fast.

User communication: tell affected users, give clear update instructions once fixed, and consider a goodwill gesture to rebuild trust.

Long term: strengthen QA for Android OS upgrades, permissions and startup paths; add automated alerting on DAU and crash-rate anomalies; and phase future rollouts so a regional canary catches this earlier.

- Q** Agreed. Thank you - we can close the interview.

WHAT WORKED · WHAT WAS MISSING

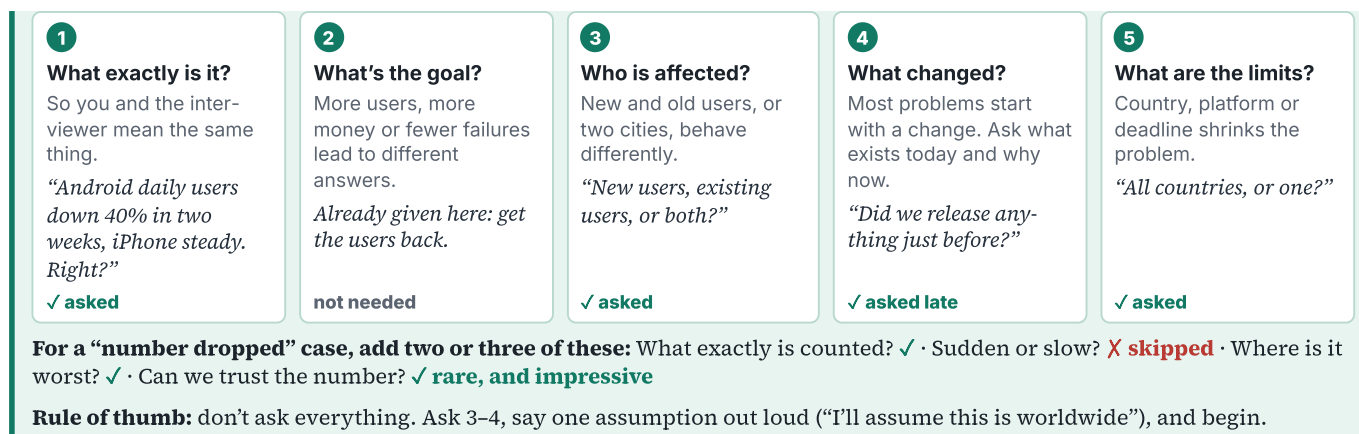
- ✓ **Checked the data first.** Asking about tracking changes and pipeline delays is rare: only 1 of the 12 root-cause cases in the IIM A and IIM B casebooks does it.
- ✓ **Pinned the definition** (unique users, at least one open a day) and checked new vs existing users.
- ✓ **Segmented by OS version and maker**, which turned "Android" into "Android 12+".
- ✓ **Closed the loose end.** The North America skew didn't fit a pure OS bug, so the candidate proposed a staged rollout and asked for proof. This is the best moment in the case.
- ✗ **Never asked "sudden or slow?"** A sudden drop usually means something broke; a slow one means users are drifting away. That one question points straight at the update and saves three turns.
- ✗ **Looked outside first**, although the drop began right after our own update.
- ✗ **Never checked the size.** Does the crash explain all 40%, or is something else going on too?
- ✗ **Forgot the business cost.** In a banking app, users who can't open it can't pay, and they call support.

CHECK · "ROLL BACK TO THE LAST STABLE VERSION" **OVERSTATED**

On Android you can't take an update back from phones that already have it. You can only stop it reaching more phones (the Play Store lets you pause a gradual release) and then send a *newer* version containing the old, working code. Teams that put risky features behind an on/off switch controlled from their own servers (a "feature flag") can simply switch the feature off.

THE SCOPING KIT · APPLIED TO A "NUMBER DROPPED" CASE

The same five questions, but in a root-cause case question 4 does the heavy lifting: most drops start with something that changed. Think of a doctor asking "did you start any new medicine?" before ordering tests.



STRONGER ANSWER



Figure 1.21 · A “metric dropped” answer in six moves. Almost any root-cause case can be answered in this order. Each turn below is tagged with its move. *Author’s summary of the RCA sequence.*

- A ① CLARIFY**
 Before I start, two quick questions. Did the drop happen suddenly, on one day, or slowly over the two weeks? And did we change anything just before it began, such as an app update?
- Q** Suddenly, the day after we released a new Android version.
Why this move: a sudden drop usually means something broke; a slow one means users are drifting away. One answer and the update is the main suspect.
- A ② RULE OUT · ③ FIND WHERE**
 Two quick checks first: is the number real, with no tracking bug, and did anything outside change, like a rival’s offer? If not, can we split Android users by their Android version, and by whether they installed the update?
- Q** The number is real and nothing outside changed. For users on Android 12 or newer who installed the update, the app crashes the moment they open it.
Why this move: clear the cheap explanations in one breath, then narrow down to exactly who is hit.
- A ④ CHECK THE SIZE**
 Does that explain the whole 40%? Say 70% of our Android users are on Android 12 or newer, and 60% of them have updated (illustrative). $70\% \times 60\% = 42\%$ who can’t open the app. That’s close to 40%, so this one bug explains the drop. North America saw it first because the update reached it first.
Why this move: if the numbers didn’t add up, there would be a second cause still to find.
- Q** Good. What do you do now, and how will you know it’s fixed?
- A ⑤ FIX · ⑥ MEASURE**
Today: stop the update reaching more phones. Send out a fixed version fast, because on Android you can’t take an update back. Text affected users to update, since they can’t open the app to read a notice.
Goal (North Star): Android daily users back to normal within a week. **Safety check (guardrail):** logins and payments still work after the rushed fix. **Trade-off:** a rushed fix is less tested, so it goes to 10% of users for a few hours before everyone. **Next time:** test every release on the newest Android versions.

FRAMEWORK · FINDING THE CAUSE: FOUR DOORS, THEN 5 WHYS

1 · The situation. A number has dropped and you have 25 minutes. The usual mistake is to grab the first idea (“a rival must have run an offer”) and spend the case defending it.

2 · An everyday example. Your scooter won’t start. You don’t open up the engine first. You check the easy things first, in an order that works just as well for an app:

Door	The question	Scooter won’t start	Android users fell 40%
1 · Data	Is the reading real?	Is the fuel gauge just broken?	Is the tracking broken?
2 · Outside	Did the world change?	Is there a petrol strike?	Did a rival launch an offer?
3 · Inside	Did we change something?	Did the mechanic work on it yesterday?	Did we release an update?
4 · Where	Where exactly is it?	Only on cold mornings?	Only on Android 12 or newer?

3 · The framework. Knock on the doors in order, because the first ones are quickest to check. If a clue points at one door, go there early: a sudden drop right after an update points at door 3. When a door opens, keep asking “why?” (the 5

Whys) until you reach something a team can fix. That is usually a missing step in a process, not a person to blame.

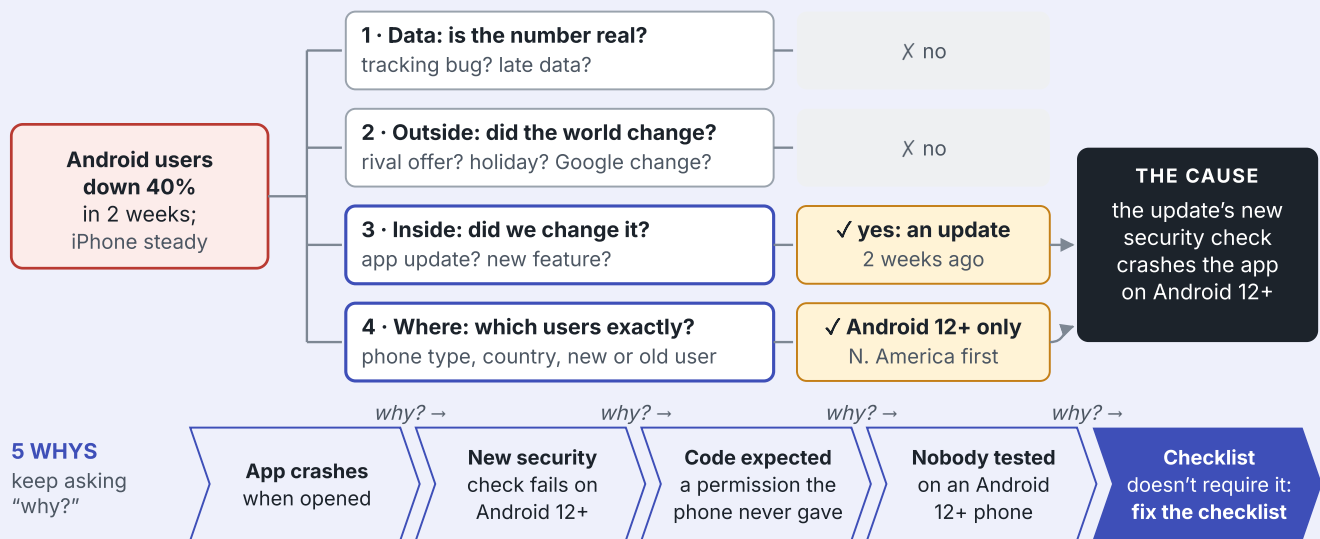


Figure 1.22 · The four doors and the 5 Whys, on this case. Two doors stay shut, two open (blue) and point to one cause. Then "why?" five times walks from the symptom to the real fix: a checklist, not a person. *Case facts from IIM A p. 308; whys 3–5 are illustrative.*

4 · When it misleads. Your scooter has a flat tyre *and* an empty tank. Fix the tyre and it still won't go. So once you find a cause, check it explains the whole drop (move ④ above). 5 Whys also goes wrong if you stop at "the developer made a mistake". People always make mistakes; the fix is the check that should have caught this one.

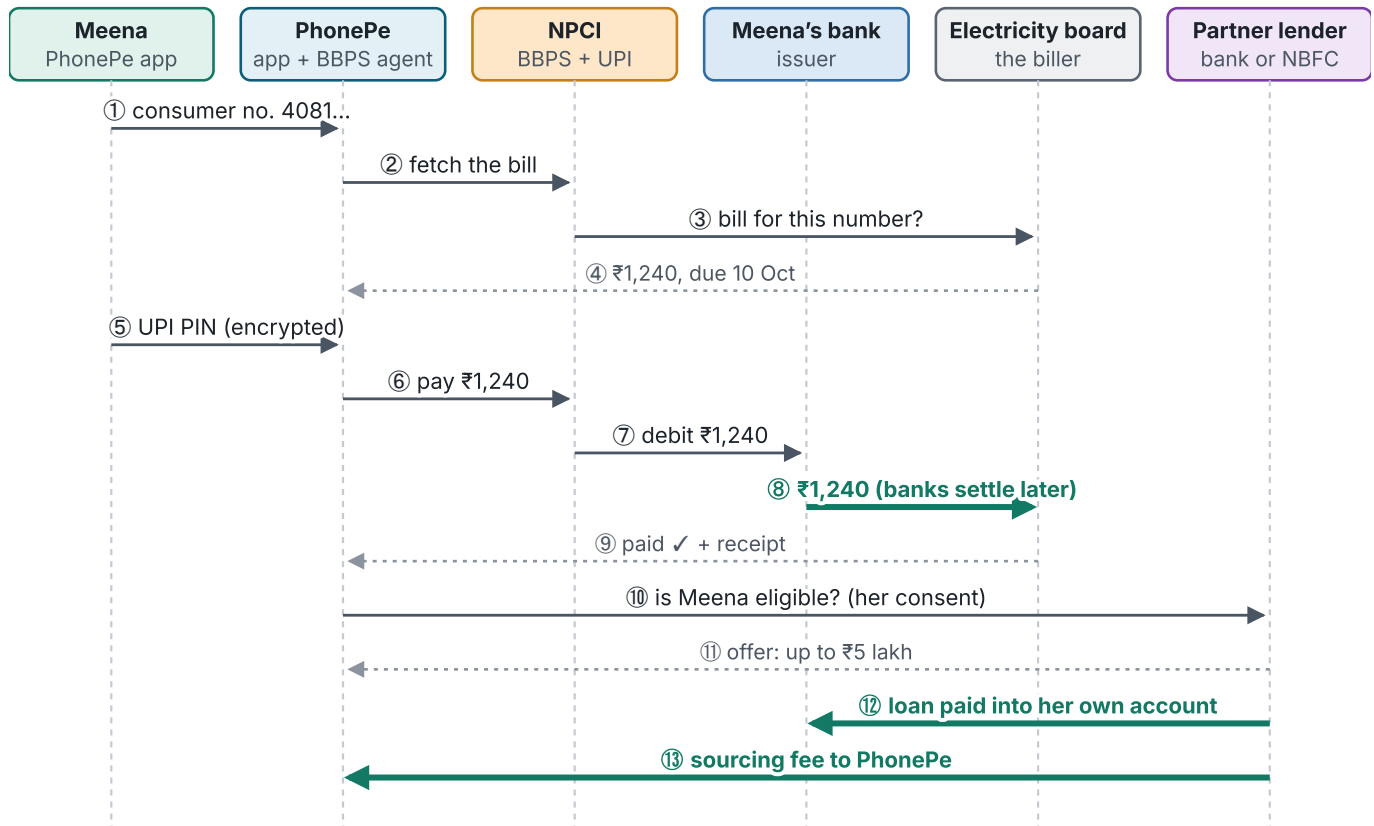
5 · In the interview. Say it once, in plain words: "First I'll check the number is real, then whether something outside changed, then what we changed, then which users are hit." Then follow the clues, and say out loud when you skip a door and why.

IN THE INTERVIEW · THE FIRST 60 SECONDS, BY CASE TYPE

Case type	Open with	The move that impresses	The usual trap
Design	Goal, who it's for, what they do today, limits (phones, law)	Checking the law or the lender first (teen credit)	Listing features before naming the user's job
Improvement	Confirm the product, pick one goal, find who is leaving	Protecting a basic before adding delight	Four personas and no decision
Metrics	Why the feature exists and for whom, then one North Star	Asking "did it cause this?" and proposing a holdout	A list of 15 equal metrics
Root cause	Exact definition, size, sudden or slow, what changed	Checking the data is real, and that the cause explains the whole drop	Blaming a competitor before checking our own release

PhonePe: free payments, paid loans

Walkthrough (illustrative persona). Meena, 41, runs a tailoring shop in Indore. On the 5th of every month she opens PhonePe, taps *Power*, and the app fetches her ₹1,240 electricity bill from her consumer number. One PIN later it's paid, and the success screen offers her a business loan. Three screens show the whole business model: a free, frequent job brings her back, and the money is made on what she sees next.



PhonePe never holds the bill money or the loan: it earns only at step ⑬.

Figure 1.23 · What happens behind Meena's three taps. Six parties, thirteen steps. PhonePe carries a message at almost every step but touches money only once, at ⑬, its own fee. Solid arrows are requests, dotted arrows replies, green arrows money. *Simplified from NPCI's Bharat BillPay and UPI flows; lender steps follow RBI (Digital Lending) Directions 2025; amounts illustrative.*

Key features, and why they're built that way

Feature	User problem	Why it's designed this way	Metric it moves
Scan & pay, P2P	Pay anyone, any shop, instantly	Scan button in the thumb zone of every screen; every transfer is free, so friends who get paid install the app	New monthly payers; payment success rate
Bill payments	Forgotten due dates, typed-in wrong amounts	Bills fetched through Bharat BillPay from the consumer number; reminders and autopay	Share of users paying a bill every month
SmartSpeaker, SmartPOD	Shopkeeper can't watch every customer's screen; fake "paid" screenshots	A speaker announces each payment aloud; SmartPOD adds card payments on the same device	Live merchants; merchant fraud complaints
Loans marketplace	Credit for people and shops banks find hard to assess	PhonePe finds the borrower and shows offers; partner banks and NBFCs lend and carry the risk	Loans per 1,000 users; 30+ days overdue
Insurance, funds, Share.Market	Small, simple investment and cover	Sold where the habit already is, in the same app, with the same KYC	Revenue per monthly transacting user

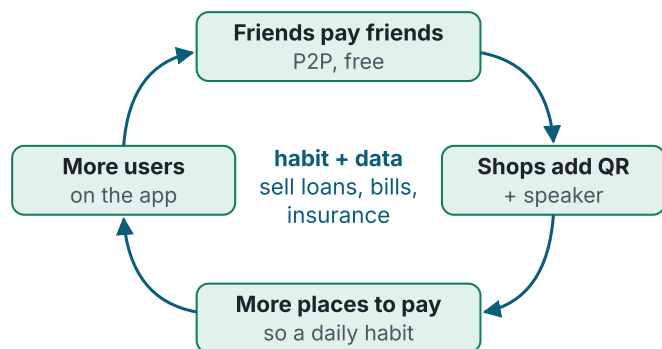


Figure 1.24 · The core loop. Each turn adds users and shops for free; money is made only at the centre, by selling to the habit. *Author's model.*

Operating revenue	₹7,920 Cr
Lending arm	₹945 Cr (2.5×)
Insurance arm	₹191 Cr
Broking arm	₹108 Cr
Govt UPI incentive	₹286 Cr

Net loss -₹2,792 Cr

Arms are subsidiaries, counted inside the total.

Figure 1.25 · PhonePe FY26, money in and out. Revenue grew 11%; the loss grew 62%, driven by stock-based pay and write-downs. *Inc42, Outlook Business, Entrackr (company filings), FY26.*

Business model in one line. Payments bring people in at a loss; revenue comes from merchant devices, bill payments, a small government incentive for UPI, and increasingly from distributing loans, insurance and investments. The lending arm alone grew from ₹378 Cr to ₹945 Cr in a year.

NORTH STAR	INPUTS	GUARDRAILS
Monthly transacting users: at least one successful payment in 30 days.	New P2P senders · live merchant QRs · payment success rate · loans per 1,000 users.	Fraud loss per ₹1 lakh moved · reversal complaints · loans 30+ days overdue.

Moat, in plain words. Habit and reach: 45.9% of UPI payments start on PhonePe (Aug 2026), and its QR codes and speakers sit in lakhs of shops. The crack in the wall: UPI makes switching free, because any UPI app scans any QR. So the moat is only as deep as the reasons to open PhonePe *first*: bills, loans, rewards, trust.

2025–26 moves. Pincode, its quick-commerce app, wound down (Dec 2025). SEBI cleared its IPO (Jan 2026); listing paused on market volatility (Mar 2026). Indus Appstore written down in FY26. Merchant lending platform and SmartPOD (speaker plus card machine). Coming: 0.4% MDR on large merchants (15 Oct 2026) and the 30% cap deadline (31 Dec 2026).

STRENGTHS

- ✓ Largest UPI share, and the default app for Bharat's shopkeepers
- ✓ Lending revenue growing 2.5× a year, on payment data rivals lack
- ✓ Trusted brand among first-time digital users

WEAKNESSES

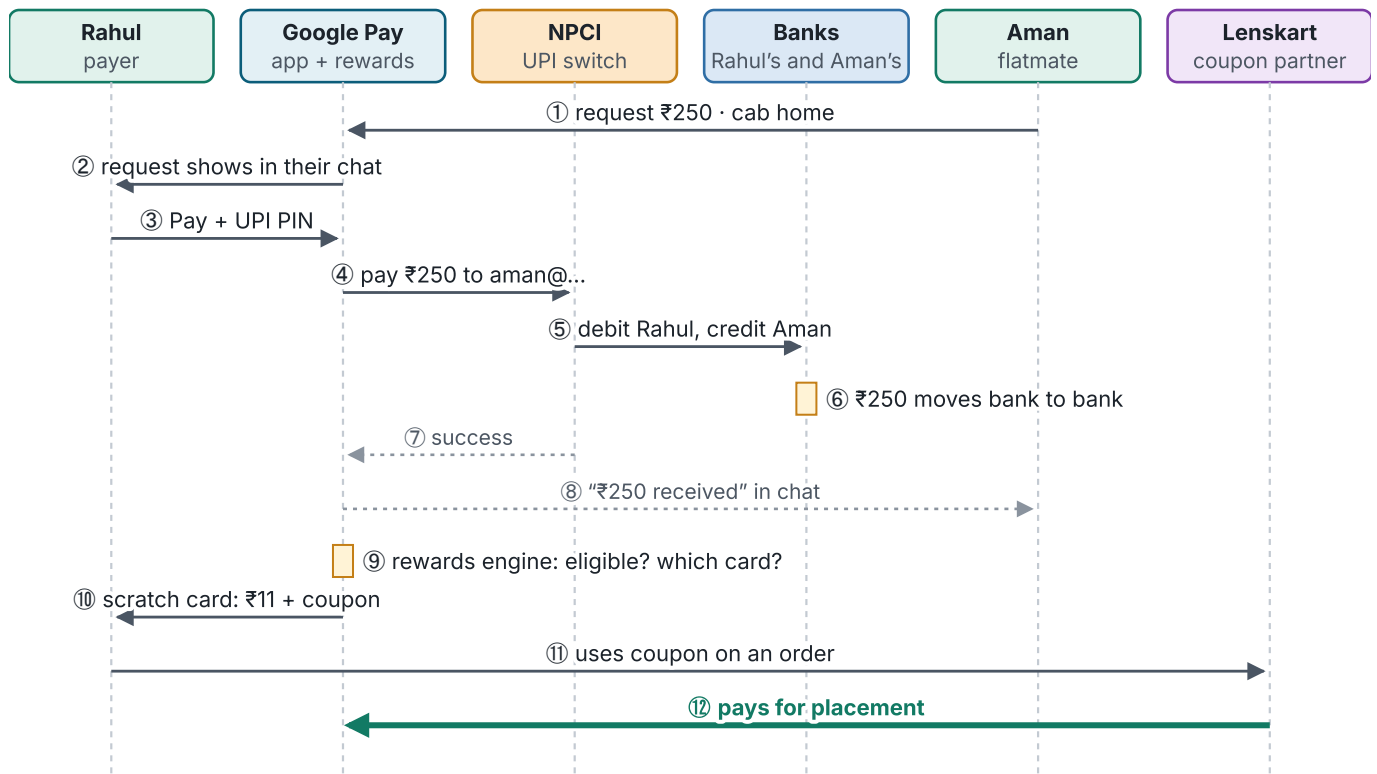
- ✗ Loss-making (–₹2,792 Cr in FY26); payments earn about nothing
- ✗ Growth could be frozen by the 30% cap
- ✗ Diversification misfires: Pincode shut, Indus Appstore written down

What I'd build next

1. A pending payment at a counter breaks trust on both sides → the speaker says “payment pending, PhonePe guarantees it” for pending payments under ₹500 → repeat payments at that shop within 7 days.
2. Small merchants pay no MDR, so they earn PhonePe nothing → use speaker takings as the credit signal for working-capital loans → approval rate and 30+ day defaults, speaker merchants vs others.
3. Large merchants like the Surat trader will push buyers to personal UPI IDs to dodge the new fee → a fee dashboard that shows the 0.4% next to the value of instant settlement, GST-ready receipts and credit built on business payments → share of large merchants keeping payments on their business QR after 15 October.

Google Pay: the payment app that lives inside Google

Walkthrough (illustrative persona). Rahul, 24, a software tester in Bengaluru, owes his flatmate Aman ₹250 for a cab. He opens Google Pay, taps Aman's face, sees their payment history as a chat, pays, and scratches a card that gives him ₹11 back and a Lenskart coupon. Google designed the whole flow to feel like messaging a friend, not visiting a bank.



Google funds the ₹11; the coupon's cost falls on Lenskart.

Figure 1.26 · Behind a ₹250 request between flatmates. The payment itself (①–⑧) is plain UPI; Google's own design sits in the chat that carries it and in the rewards engine at ⑨, which a partner brand helps pay for at ⑫. *Simplified from NPCI UPI collect-request flow; names and amounts illustrative, Sep 2026.*

Key features, and why they're built that way

Feature	User problem	Why it's designed this way	Metric it moves
People-first home, chat threads	"Did I pay you? How much do I owe?"	Every payment sits in a thread with that person, like WhatsApp; requests and payments go both ways	P2P payments per user; repeat pairs
Scratch cards, partner offers	Payments are boring, and every app is the same	A variable reward after eligible payments; partner brands fund many of the prizes	Payments per user; coupon redemption
Pocket Money (UPI Circle)	Parents want children to pay digitally, within limits	Built on NPCI's delegated payments: up to ₹15,000 a month, or approve each payment (Dec 2025)	Family circles created; child payments
Flex card, loans	Credit for young earners with thin credit files	A RuPay credit card with Axis Bank (Dec 2025), usable on UPI; personal loans from partners such as DMI Finance	Cards issued, loans disbursed, overdue rate
Scam alerts	"Customer care" callers who ask you to share your screen	A warning when a payment is made during screen sharing (Android 11+); Google says it showed 4.1 crore warnings from Oct 2024 to Dec 2025	Fraud reports per lakh payments

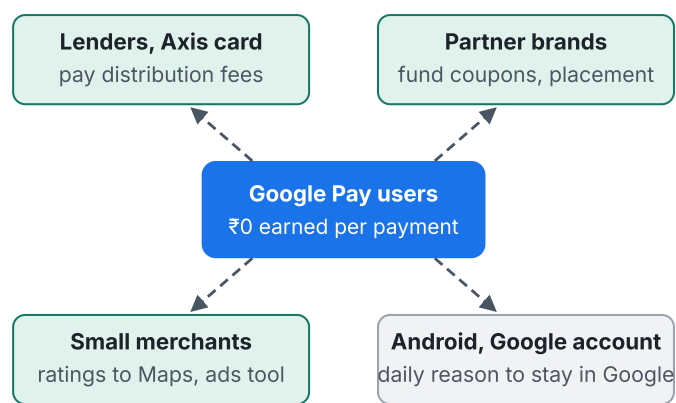
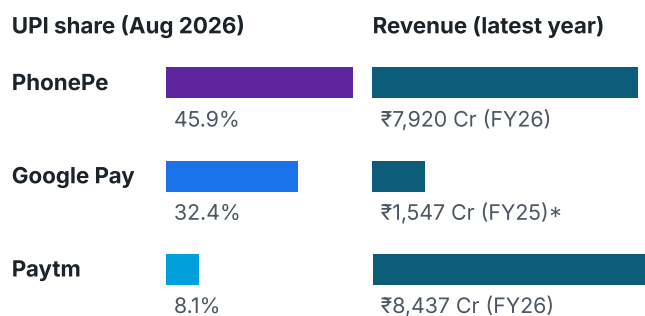


Figure 1.27 · How a free payments app earns for Google. Green boxes pay in rupees; the grey one pays in strategy, keeping Google in users' daily lives. *Author's model from Google announcements, Dec 2025.*



*Indian entity only; Google books some value elsewhere.
Share of payments ≠ share of money.

Figure 1.28 · UPI share and revenue barely line up. Paytm, with a quarter of Google Pay's share, reports over five times its Indian revenue, because it sells devices and loans to merchants. *NPCI via Inc42 (Aug 2026); company filings via Storyboard18 (Feb 2026), Paytm IR (May 2026), Inc42 (FY26).*

Business model in one line. Google Pay India's own revenue is small (₹1,547 Cr in FY25, profit ₹50 Cr) and flat for five years. It earns from loan and card distribution, partner-funded offers and fees on card bill payments (0.5–1% plus GST when paid by card; free from a bank account by UPI). Its bigger value is strategic: payments are the most frequent reason to open a Google app in India.

NORTH STAR	INPUTS	GUARDRAILS
Monthly payers making 10+ payments: a habit measure, not a download count.	P2P pairs · payment success rate · reward redemption · Flex cards issued.	Scam reports per lakh payments · reward cost per payment · overdue card balances.

Moat, in plain words. It comes pre-installed in trust: a Google account, Android, and a clean interface people already know. Chat threads make leaving costly in a small way, because your payment history with each friend lives there. Its fraud tools draw on Google's scale. **2025–26 moves:** Flex RuPay credit card with Axis Bank and Pocket Money on UPI Circle (both Dec 2025); screen-sharing scam alerts; merchant ratings synced to Google Maps; convenience fees on card-funded bill payments.

STRENGTHS

- ✓ 32.4% of UPI; the cleanest, most social interface
- ✓ Google's trust and scale in fraud detection
- ✓ Rewards partly funded by partner brands

WEAKNESSES

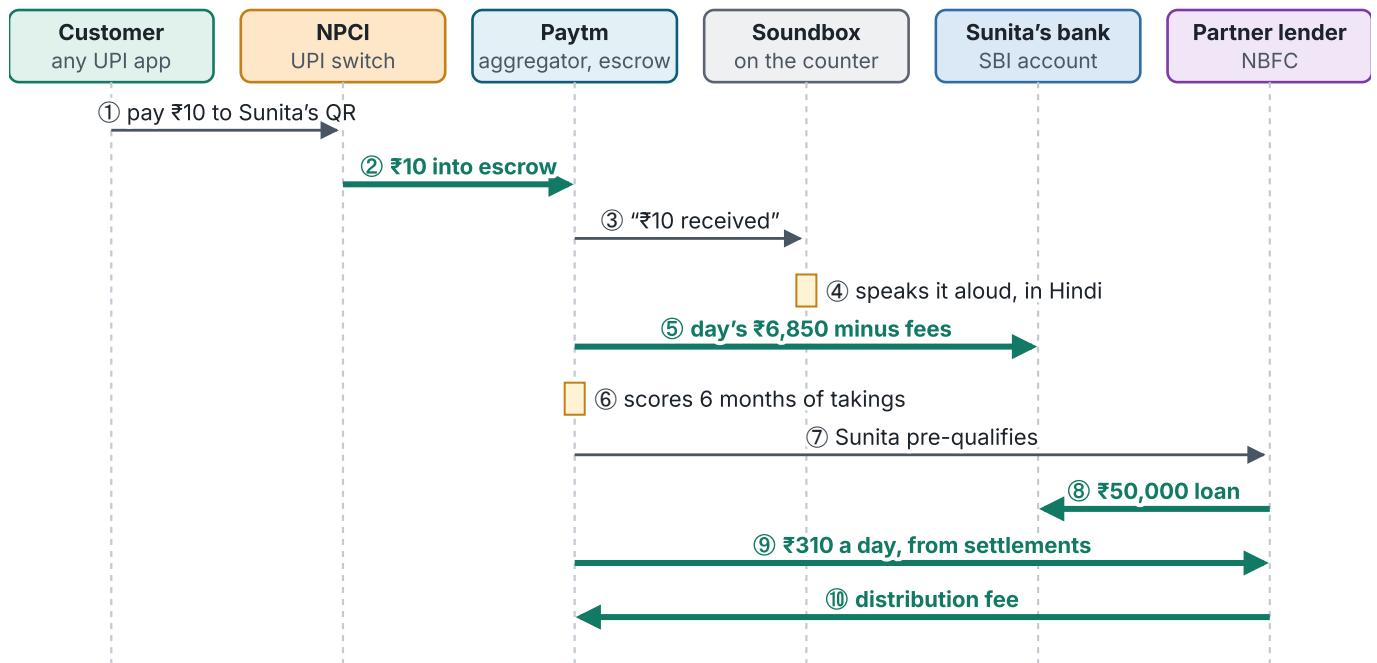
- ✗ Small Indian revenue for its size; every financial product depends on partners
- ✗ Weaker with offline merchants than PhonePe and Paytm
- ✗ Also above the 30% cap, if it is ever enforced

What I'd build next

1. Group expenses (flats, trips) are tracked in spreadsheets and chased by message → shared groups with running balances and one-tap "settle up" by UPI → active groups settling at least monthly.
2. Most scams happen while the victim is on a call with the scammer → with permission, warn when a payment to a new payee starts during a call from an unsaved number → scam reports per lakh payments, and genuine payments abandoned after the warning.
3. Flex cardholders are new to credit and may overspend → a weekly "what you owe" nudge in the chat-style home, with autopay from the bank account → share of card bills paid in full on time.

Paytm, and the super-app question

Walkthrough (illustrative persona). Sunita, 45, runs a tea stall outside Lucknow's Charbagh station. Her Paytm Soundbox calls out every payment, including the ₹10 ones, so she never checks a customer's phone. At night the day's ₹6,850 lands in her SBI account, and the app offers her a ₹50,000 loan repaid in small daily amounts. Paytm lost most of its consumer lead to PhonePe and Google Pay, then rebuilt around shopkeepers like Sunita.



Money in escrow is the customers' and Sunita's, never Paytm's own; Paytm earns device fees, card fees and step ⑩.

Figure 1.29 · Behind a ₹10 chai payment, and the loan it leads to. As a payment aggregator, Paytm receives customers' money in an escrow account and settles it to Sunita each evening (⑤). Her daily takings become the credit signal (⑥), and loan repayments come out of future settlements (⑨). *Simplified from RBI payment-aggregator rules and Paytm's merchant-lending model; amounts and loan terms illustrative.*

Key features, and why they're built that way

Feature	User problem	Why it's designed this way	Metric it moves
Soundbox, card machine	Can't check every customer's screen at a rush-hour counter	A device that speaks each payment aloud; sold as a monthly subscription, and the card machine accepts UPI and cards	Subscription merchants (1.51 crore, Q4 FY26)
Merchant loans	No formal income proof, so banks say no	Partner lenders use Paytm's payment history; repayments taken in small amounts from daily settlements	Loans disbursed; collection rate
Consumer UPI	Pay anywhere, keep the Paytm habit	After its own bank was shut, Paytm moved users to four partner-bank handles (Axis, HDFC, SBI, Yes)	UPI share (8.1%, Aug 2026)
Financial services distribution	Loans, insurance and equities in one known app	Paytm earns a fee per product sold; lenders keep the loans on their books	Revenue: ₹750 Cr in Q4 FY26, +38%
Paytm Intelligence (Sep 2026)	Banks and insurers want AI for fraud, credit and claims	Sells AI agents built on Paytm's payment data and models to financial firms in India and the UAE	New: enterprise contracts

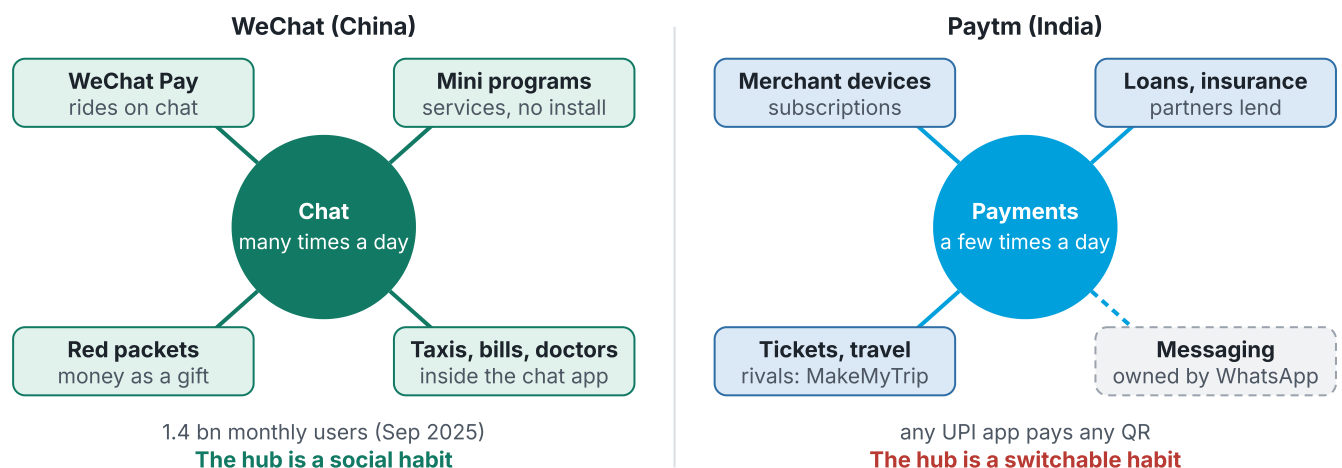


Figure 1.30 · Why WeChat became a super-app and Paytm did not. WeChat's hub is conversation, which people return to many times a day and can't take elsewhere; its services ride on that. Paytm's hub is payments, which UPI makes interchangeable, and the social hub in India belongs to WhatsApp. *Tencent 2025 annual results (MAU as of 30 Sep 2025); author's model.*

CHECK · "PAYTM IS INDIA'S SUPER-APP" **OVERSTATED**

Paytm tried: shopping, games, travel, movies and a bank, all in one app. Three things worked against it. In India, messaging belongs to WhatsApp, so no payments app owns the daily social habit that carried WeChat Pay. Android and the Play Store make standalone apps easy to install, so mini-apps save users little. And UPI lets every app pay every shop, so payments can't lock users in. Paytm today is a merchant-payments and financial-distribution company with some consumer services, and its results improved when it narrowed.

Revenue (₹ Cr)



Profit after tax (₹ Cr)



Q1 FY27: revenue ₹2,448 Cr (+28%), profit ₹220 Cr

NORTH STAR

Merchants on a paid device subscription who transact daily.

INPUTS

Devices deployed · merchant payment volume (GMV) · loans distributed per 100 merchants.

GUARDRAILS

Device churn · loan collection rate · compliance findings.

Figure 1.31 · From loss to profit in one year. Revenue rose 22% and EBITDA improved by ₹2,008 Cr, to ₹502 Cr. *Paytm investor relations, 6 May 2026; Q1 FY27 results, Jul 2026.*

Moat, in plain words. A field force and 1.51 crore merchants paying for devices: switching means returning a device and retraining staff, which a UPI app can't replicate with software alone. **2025–26 moves:** first full year of profit (FY26); RBI cancelled the Payments Bank licence (24 Apr 2026), ending a two-year overhang; Paytm Intelligence, AI agents for banks and insurers (Sep 2026).

STRENGTHS

- ✓ Earns from merchants: devices, card fees, loan distribution
- ✓ Profitable, with revenue growing faster than costs
- ✓ Deep merchant data for credit and fraud models

WEAKNESSES

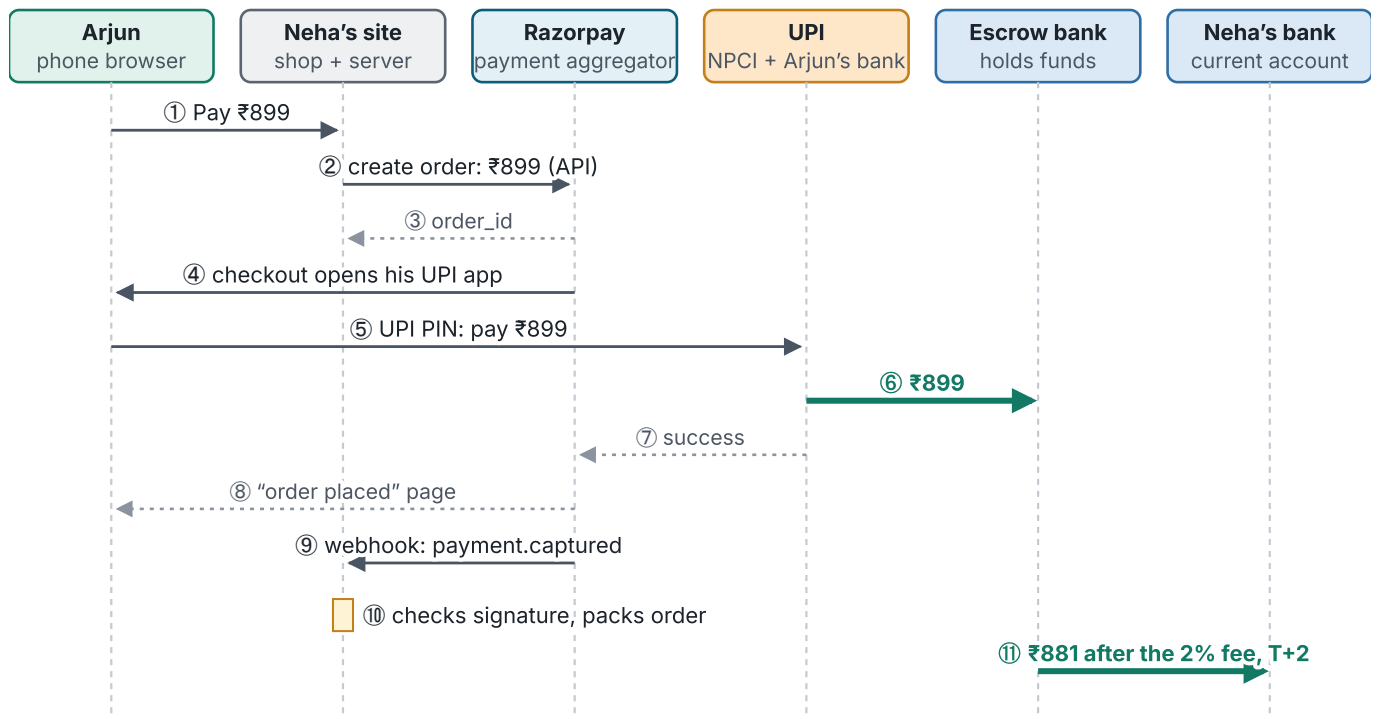
- ✗ Consumer UPI share fell to 8.1% after the 2024 curbs
- ✗ Regulatory history makes RBI and partners cautious
- ✗ AI pivot is new and unproven as a business

What I'd build next

1. Soundbox merchants borrow informally at high rates when stock runs out → a small credit line that grows with daily takings, drawn by voice on the Soundbox ("bolo, ₹5,000 chahiye") → credit-line use and 30+ day defaults per 100 merchants.
2. Shopkeepers can't read dashboards while serving customers → the Soundbox reads out the day's total at closing time in their language → weekly active merchants on the app, and device churn.

Razorpay: the plumbing behind India's online checkouts

Walkthrough (illustrative personas). Neha sells skincare from her own website. Arjun, a customer in Jaipur, buys a ₹899 serum on his phone. The checkout sheet, the jump to his UPI app and the "order placed" page are all Razorpay, though Arjun never sees the name. Neha sees it every evening, on the dashboard where her day's payments, failures and settlements live. Razorpay's customer is Neha; Arjun's experience is its product.



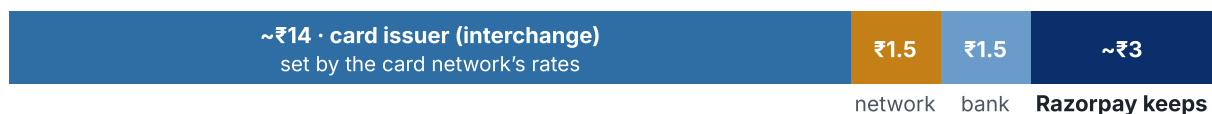
Razorpay keeps about ₹18 here; buyers' money waits in an escrow account, not Razorpay's own.

Figure 1.32 · Behind one online order. Neha's server talks to Razorpay through an API (②–③) and hears about the payment through a webhook (⑨): two ideas every payments PM uses daily. The money waits in an escrow account and reaches Neha, minus the fee, two working days later (⑪). Simplified from Razorpay's public API and webhook docs and RBI payment-aggregator rules; fee from razorpay.com/pricing, Sep 2026.

Key features, and why they're built that way

Feature	User problem (the merchant's)	Why it's designed this way	Metric it moves
Checkout	Every payment method needs its own bank contract and code	One integration for UPI, cards, netbanking, EMI and wallets; saved cards as tokens, as RBI requires	Checkout conversion; success rate
Payment links, pages	Instagram sellers and tutors have no website	A link sent on WhatsApp opens a hosted payment page	New merchants live in 7 days
Subscriptions	Chasing monthly fees by hand	UPI Autopay and card mandates debit customers on schedule	Renewal rate
RazorpayX, Payroll	Paying vendors and staff means bank portals and spreadsheets	A current account with payouts and payroll run by API; 50,000+ businesses (company figure)	Products per merchant
Agentic payments	AI assistants that shop need safe ways to pay	UPI Reserve Pay: the user authorises once, within set limits, and an app or AI agent completes several debits; pilots with NPCI and OpenAI, launches on Claude and Codex (2025–26)	Agent-initiated payment volume; disputes

Where a merchant's ₹20 goes on a ₹1,000 card payment (standard 2% plan)



Headline fee vs what Razorpay earns on all the money it moves



Figure 1.33 · A gateway keeps a thin slice. Top: on card payments, most of the fee passes through to banks. Bottom: across all its volume, Razorpay earns roughly a tenth of its list price. That is why it keeps adding products further from the payment. Split illustrative; list price: razorpay.com/pricing (Sep 2026); revenue FY25 (Inc42); volume is the company's 2026 annualised claim, so the ratio is rough.

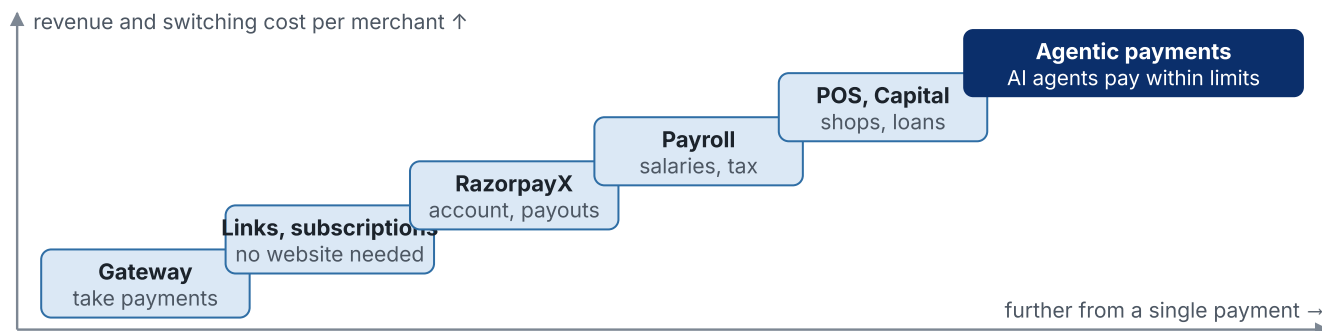


Figure 1.34 · Razorpay's product ladder. Each step away from the single payment earns more per merchant and is harder to switch away from. A merchant can change gateways in weeks; changing payroll and bank accounts takes months. *Author's model from Razorpay product pages, 2026.*

NORTH STAR Successful payment value from merchants active this month.	INPUTS Payment success rate · new merchants live in 7 days · products per merchant.	GUARDRAILS Fraud and chargebacks · settlement delays · minutes of downtime.
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Business model in one line. A fee on each payment (2% on the standard plan, much less for large merchants), plus banking, payroll and lending products that earn more per merchant. FY25 revenue was ₹3,783 Cr (+65%), with a net loss of ₹1,209 Cr, mostly stock-based pay and a one-off tax on moving its holding company back to India. **Moat, in plain words:** being wired into thousands of merchants' systems (checkout code, webhooks, reconciliation, payroll). Leaving means rebuilding that wiring. **2025–26 moves:** the move back to India (May 2025); a confidential IPO filing (12 Jun 2026); agentic-payments launches with NPCI, OpenAI and Anthropic.

STRENGTHS

- ✓ Developer-first products that other fintechs copy
- ✓ A ladder of products with rising switching costs
- ✓ Early in AI-agent payments

WEAKNESSES

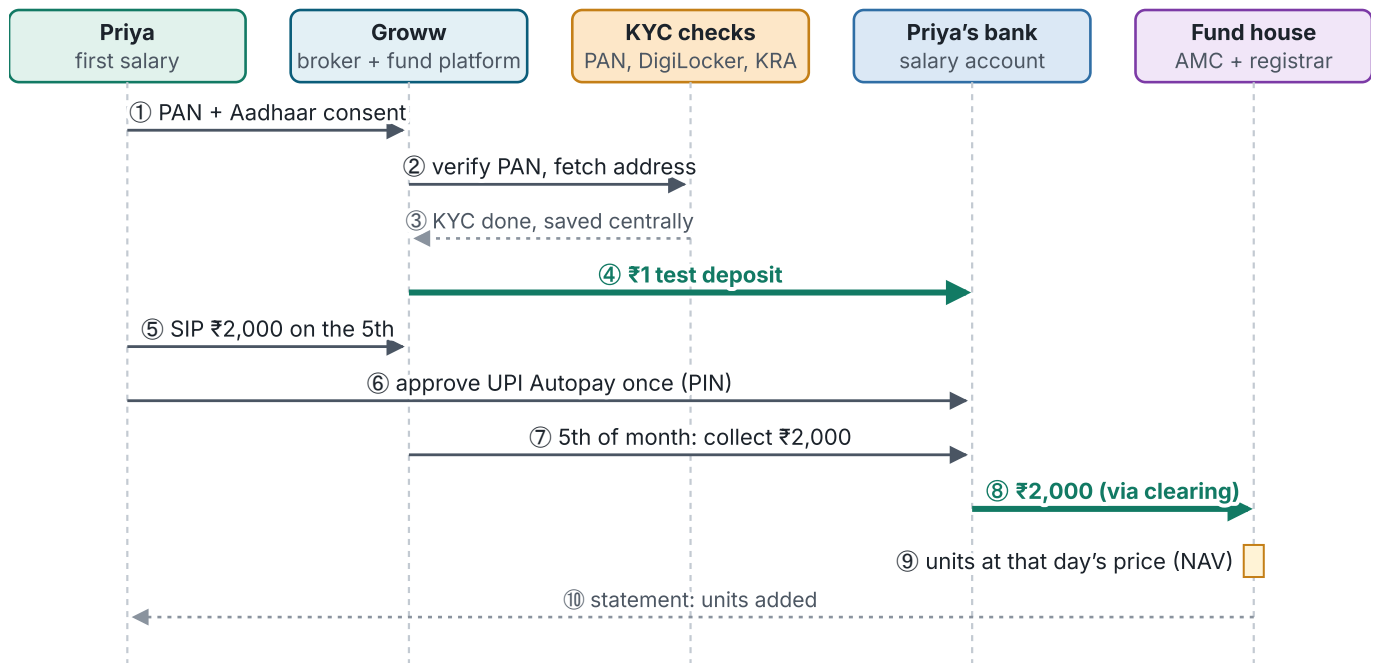
- ✗ Thin gateway margins; zero-MDR UPI and rivals (PayU, Cashfree, Juspay) squeeze price
- ✗ Net loss in FY25; IPO timing depends on showing profit
- ✗ Every product rests on bank partners and RBI licences

What I'd build next

1. Neha loses orders every time a card payment fails → when a card fails, offer UPI in the same sheet with the amount pre-filled → orders recovered per 100 failed payments.
2. Small D2C brands run short of cash before a sale season → advances repaid as a fixed share of daily settlements, priced on their Razorpay history → advances taken and loss rate.
3. Merchants fear AI agents placing wrong or duplicate orders → per-agent spending caps and an "agent orders" view on the dashboard → disputes per 1,000 agent payments.

Groww vs Zerodha vs smallcase: three ways to earn from investors

Walkthrough (illustrative personas). Priya, 23, gets her first salary in Pune and wants to "start investing". On Groww she finishes KYC in minutes: a PAN check, her address pulled through DigiLocker, and a ₹1 test deposit to her bank account. She picks a Nifty 50 index fund and sets a ₹2,000 SIP (a fixed monthly investment) on the 5th, paid by UPI Autopay. Karan, 35, is the opposite user: he trades options on Zerodha's Kite on a laptop, many times a day. A year later Priya buys a themed "basket" of 12 stocks through smallcase, inside her broker's app, for a ₹100 fee. Three companies, three very different ways of making money from the same investors.



Direct plan: the fund pays Groww no commission, so this SIP earns Groww almost nothing.

Figure 1.35 · Behind Priya's first SIP. Groww does the KYC, sets up the mandate and routes the money, but a direct-plan SIP pays it nothing: the product is a doorway to stocks, F&O and margin trading. *Simplified from SEBI KYC (KRA) rules and NPCI UPI Autopay; money routing via exchange clearing simplified; amounts illustrative.*

Three business models side by side

	Groww	Zerodha	smallcase
Built for	First-time investors on phones	Active traders and serious investors	Investors who want a ready-made portfolio
Front door	Direct mutual funds, SIPs from ₹100	Low, flat brokerage; Kite trading screen; free lessons (Varsity)	Themed baskets by SEBI-registered advisers, inside your broker's app
Earns from	Brokerage on stocks and F&O, margin-trading interest	Brokerage (mostly F&O), margin-trading interest	₹100 + GST per basket purchase (capped at 1.5%), ₹10 per SIP instalment
Latest numbers	1.33 crore active clients, 29% of NSE's (Aug 2026). FY26 revenue ₹4,645 Cr, profit ₹2,083 Cr	~68 lakh active clients, ~15% (Jun 2026). FY26 revenue ~₹8,500 Cr (flat), profit ~₹4,300 Cr	FY25 revenue ₹106 Cr (+57%), net loss ₹34 Cr
Moat	Simplest onboarding; the brand first-time investors trust	Trust, low cost, trader tools; profitable without outside funding	Embedded in many brokers, so brokers find it costly to drop

Per active client, FY26 (₹)

revenue

Zerodha ~12,500

Groww ~3,500

profit

Zerodha ~6,300

Groww ~1,560

Groww has 2× the clients; Zerodha earns ~4× per client.

Figure 1.36 · Breadth vs intensity. Groww earns from many light users, Zerodha from fewer heavy traders. *FY26 revenue and profit (company results, Entrackr, Aug 2026) ÷ mid-2026 NSE active clients; approximate.*

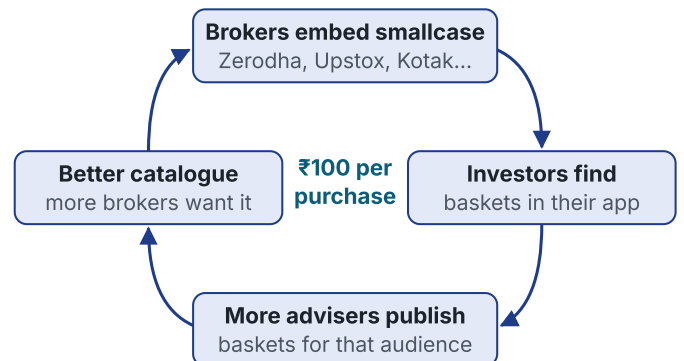


Figure 1.37 · smallcase grows through other companies' apps. Brokers bring the investors, advisers bring the baskets, and each side makes the other more valuable. *Author's model; fee from smallcase.com, 2026.*

REAL-LIFE EXAMPLE · WHAT SEBI'S F&O RULES DID TO BOTH BROKERS

In September 2024 SEBI found that 93% of individual futures-and-options traders lost money over FY22–FY24. From late 2024 it made contracts larger, limited weekly expiries, and made exchanges charge all brokers the same fees, which ended the rebates brokers used to earn. Zerodha, built on active traders, felt it most: FY26 brokerage income fell about 11% and its exchange-rebate income went to zero, though profit held at about ₹4,300 Cr thanks to margin-trading interest. Groww, with a broader base of small investors, kept growing: Q1 FY27 revenue was up 66% and profit ₹735 Cr.

	North Star	Inputs	Guardrails
Groww	Investors who invest in 3+ of the last 6 months	KYC completion, SIPs live at 12 months, stock and F&O activations	Share of new users trading F&O in their first 90 days; complaints
Zerodha	Active clients with assets held, not just trades	Assets per client, margin-trading book, Kite uptime	Order failures on busy days; loss-making F&O streaks
smallcase	Monthly investors in at least one basket	Broker integrations, baskets per investor, basket-SIP continuation	Baskets underperforming their benchmark; mis-selling complaints

2025–26 moves. Groww listed on the stock exchanges (Nov 2025) and posted a profit of ₹2,083 Cr in its first year as a listed company. Zerodha's FY26 was flat, with margin-trading interest now about a tenth of revenue. smallcase crossed ₹100 Cr of revenue in FY25 and narrowed its losses.

STRENGTHS

- ✓ **Groww:** the easiest first step for millions of new investors
- ✓ **Zerodha:** the most profit per client, with no outside funding
- ✓ **smallcase:** distribution through other brokers' apps, at almost no acquisition cost

WEAKNESSES

- ✗ **Groww:** earns most on the riskiest products it sells to beginners
- ✗ **Zerodha:** revenue tied to trading intensity, which regulators are cooling
- ✗ **smallcase:** small revenue; depends on brokers keeping it embedded

What I'd build next

- 1. Groww.** Many first SIPs stop within a year, often after the first market fall → a “stay the course” screen after a dip that shows her own SIP's long-run average cost → SIPs still live at 12 months.
- 2. Zerodha.** Most F&O traders lose money, and losses often come in streaks → an optional cooling-off: after three losing weeks, suggest a one-week pause on option buying, with the trader's own numbers → share of traders who take the pause, and their 6-month outcome against similar traders who don't.
- 3. smallcase.** Investors buy a basket once and forget it → basket SIPs tied to a named goal (“Home deposit, 2030”) with progress shown in the broker's app → basket-SIP continuation at 12 months.

The five teardowns side by side

Product	Job for its user	How it earns	Latest numbers	North Star	Biggest risk (2026)
PhonePe	Pay anyone, any shop, any bill	Loans, insurance and broking sold to the payment habit; merchant devices	45.9% of UPI; FY26 revenue ₹7,920 Cr, loss ₹2,792 Cr	Monthly transacting users	30% cap freezing new users
Google Pay	Pay friends and shops, simply	Loan and card distribution, partner offers, card-bill fees	32.4% of UPI; Indian entity FY25 revenue ₹1,547 Cr	Payers with 10+ payments a month	Small revenue; dependence on partners
Paytm	Help a shop accept money and get credit	Device subscriptions, card fees, loan distribution	8.1% of UPI; FY26 revenue ₹8,437 Cr, profit ₹552 Cr	Daily-active subscribed merchants	Regulatory trust; unproven AI pivot
Razorpay	Let a business take and move money online	Fee per payment; banking, payroll, lending	FY25 revenue ₹3,783 Cr, loss ₹1,209 Cr	Payment value from active merchants	Price war; thin gateway margins
Groww	Make a first investment easy	Brokerage, F&O, margin interest	1.33 crore active clients; FY26 profit ₹2,083 Cr	Investors active 3 of 6 months	Earning most from beginners' riskiest trades
Zerodha	Trade cheaply with good tools	Brokerage, margin interest	~68 lakh clients; FY26 profit ~₹4,300 Cr	Active clients holding assets	F&O curbs cutting trading intensity
smallcase	Buy a ready-made portfolio	₹100 per basket purchase	FY25 revenue ₹106 Cr, loss ₹34 Cr	Monthly basket investors	Brokers dropping or copying it

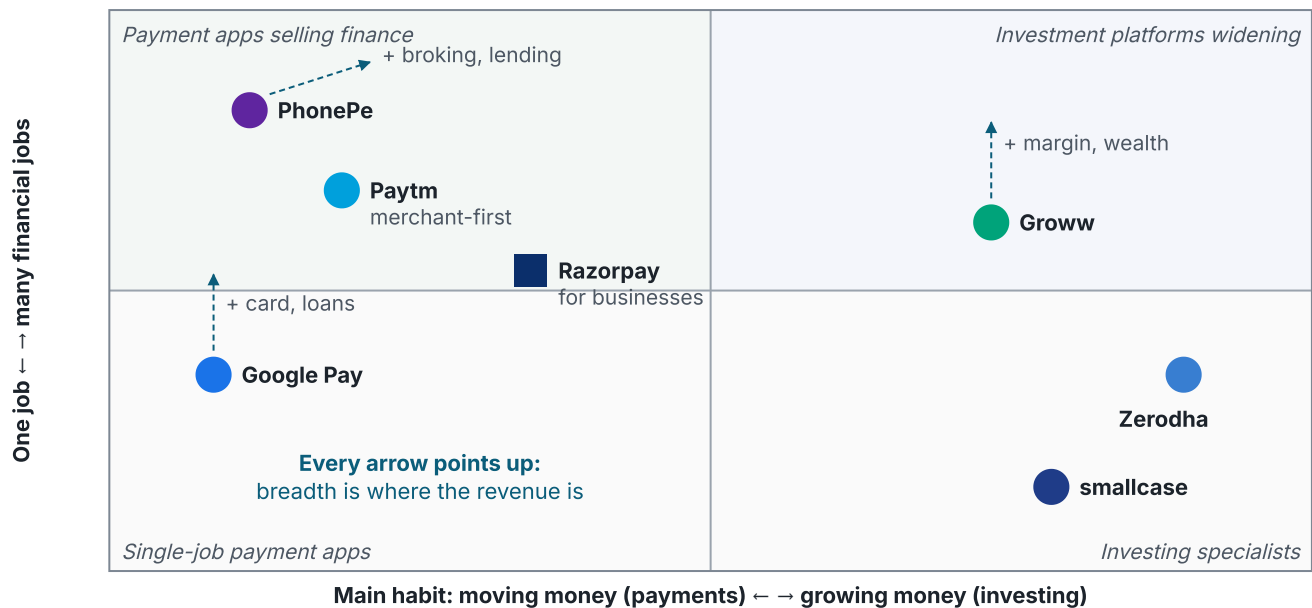


Figure 1.38 · Where the five sit, and where they're heading. Across: the habit each product owns. Up: how many financial jobs it does. Payment apps add credit and investing; investing apps add lending against shares. Nobody moves down, because single-job products earn too little in India's zero-fee payments market. *Author's placement, Sep 2026; Razorpay (square) serves businesses, so it is placed by breadth of business products.*

IN THE INTERVIEW · "WHICH OF THESE WOULD YOU RATHER BE PM OF?"

Pick one and argue from the business model, not the app's looks: "Paytm. It is the only one here earning real money from merchants rather than from the payment itself, it has just turned profitable, and the next product decision matters: whether merchant credit or the new AI business gets the best engineers. That's a PM's job with real trade-offs." Then name the risk you'd watch.

KEY IDEA · READ ANY FINTECH IN THREE NUMBERS

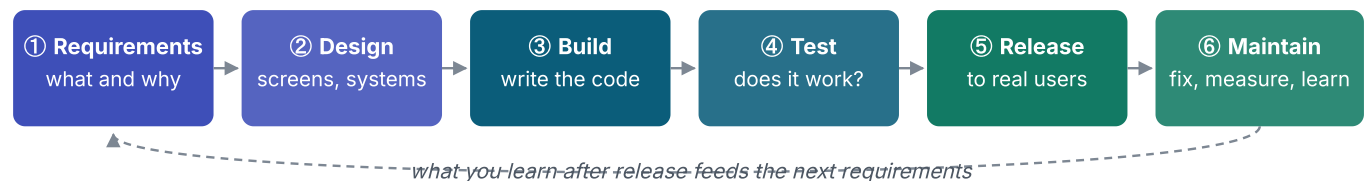
Share (how many use it) · **revenue per user** (what the habit earns) · **loss or default rate** (what it costs). PhonePe leads the first, Zerodha the second, and every lender lives or dies by the third.

How software is built: from an idea to an app update

September 2026, PhonePe's merchant team has two jobs on its board. The first has a date nobody can move: from 15 October, large merchants pay 0.4% on UPI payments above ₹2,000, so fee calculation, invoices and settlement reports must be live on day one. The second is open-ended: "help merchants like Hitesh keep payments on their business QR", where nobody yet knows what will work. Same team, same engineers, and two very different ways to plan the work.

The software development life cycle (SDLC)

Every piece of software, from a button to a bank, goes through the same six stages. The **SDLC** is simply their names. What changes between teams is how long each stage lasts and how often they loop.



What the PM owns at each stage

PRD, user stories, success metric	review flows with designer, engineers	answer questions, cut scope if late	acceptance criteria: what "done" means	go / no-go, roll-out plan, comms	watch metrics, prioritise fixes
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Figure 1.39 · Six stages, and the PM's job in each. The PM writes little code, but owns the "why" at the start, the definition of "done" in the middle, and the learning at the end. *Stages as in the IIM A Product Mindset tech curriculum (p. 70); PM roles: author's summary.*

PRD (product requirements document): the one document that says what is being built, for whom, why, and how success is measured.

User story: one requirement from the user's side: "As a merchant, I want to see the fee on each payment, so that I can check my settlement."

Acceptance criteria: the tests a story must pass to count as done: "Fee shown to the paisa; zero on payments of ₹2,000 or less."

MVP (minimum viable product): the smallest version that tests whether the idea works with real users.

Waterfall and Agile: two ways to walk the stages

Waterfall walks the six stages once, in order: all requirements, then all design, then all the code, then all the testing. **Agile** walks all six in a small loop every two weeks or so, shipping a working slice each time and learning from users before the next slice. Think of building a house (you can't pour the foundation after the roof) against cooking for guests (you taste and adjust as you go).

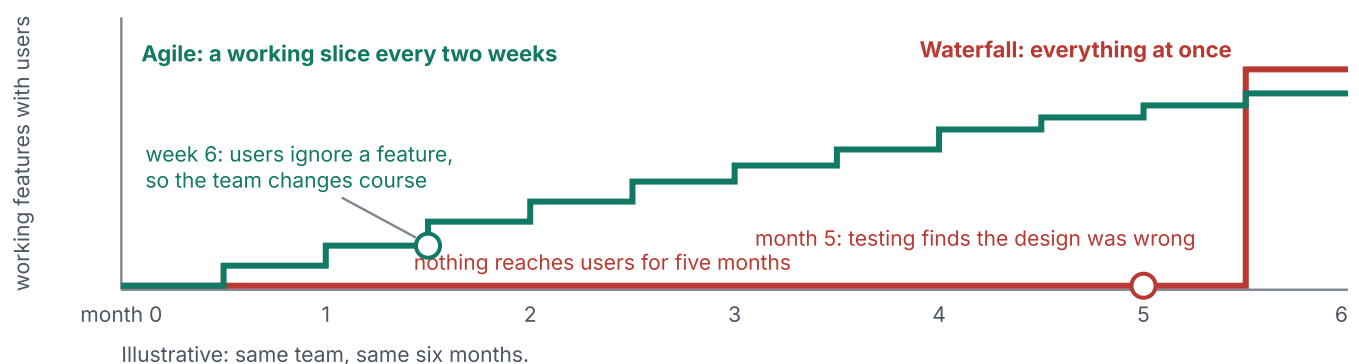


Figure 1.40 · When users see the work, and when mistakes surface. Waterfall can finish at the same time, but its first real feedback arrives at the end, when changes cost the most. Agile trades a tidy plan for early learning. *Illustrative; comparison after IIM A Product Mindset (pp. 71–74).*

Use	When the work looks like...	Payments example
Waterfall-style plan	Requirements are fixed and known, the date can't move, and a mistake is expensive	The 15 Oct MDR change; RBI's card-tokenisation deadline (1 Oct 2022)
Agile	You don't yet know what users will value, and learning fast beats a perfect plan	"Keep merchants on their business QR"; a new rewards format
Mix (most real teams)	A fixed core with open edges	Fee engine planned to the day; merchant fee dashboard iterated in sprints

Scrum and Kanban: the two common ways to run Agile

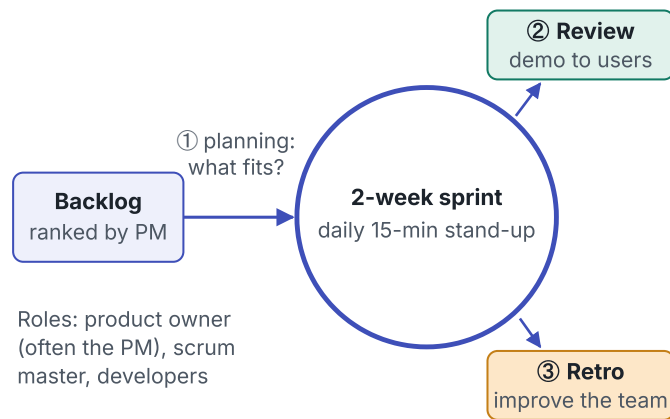


Figure 1.41 · Scrum: fixed-length loops. Work is chosen at the start of each sprint and demoed at the end. *After the Scrum Guide (2020).*

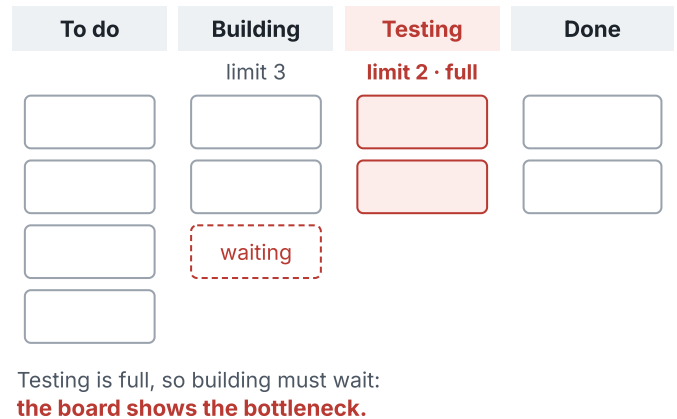


Figure 1.42 · Kanban: continuous flow with limits. Each column caps work in progress; a full column is a bottleneck to fix. *After Toyota's kanban and David J. Anderson's software Kanban.*

Scrum suits a team building new features on a steady rhythm. **Kanban** suits work that arrives unpredictably, such as payment-failure fixes or merchant support escalations. A **WIP limit** (work in progress) caps how many tasks can sit in a column, which stops a team from starting ten things and finishing none.

Light maths: when will it ship?

Teams size stories in **story points** (relative effort, not hours) and track **velocity**, the points finished per sprint. Say the merchant fee dashboard is 120 points and the last three sprints delivered 18, 22 and 20. At the average of 20, that is $120 \div 20 = 6$ sprints, or 12 weeks. But velocity wobbles: at 22 it is 5.5 sprints, at 18 it is 6.7. So the honest answer is **"11 to 14 weeks"**, not a date. If the business needs 8 weeks, the PM's lever is scope: cut the dashboard to 80 points ($80 \div 20 = 4$ sprints) by dropping charts and keeping the fee table.

WHEN IT BREAKS

- **Agile in name only:** daily stand-ups, but releases once a quarter. Nothing is learned between them.
- **Points as a productivity target:** tell teams to raise velocity and estimates quietly inflate, so velocity rises while output doesn't.
- **Fixed deadlines in an open backlog:** regulatory work treated as "we'll pick it up next sprint" misses the date. Plan it backwards from the deadline.

KEY IDEA · THE PM DECISIONS THIS DRIVES

- Which method fits which project (fixed vs open).
- What is in each sprint, ranked in the backlog.
- What "done" means: acceptance criteria, not "the code is written".
- When late, cut scope, not testing.

IN THE INTERVIEW

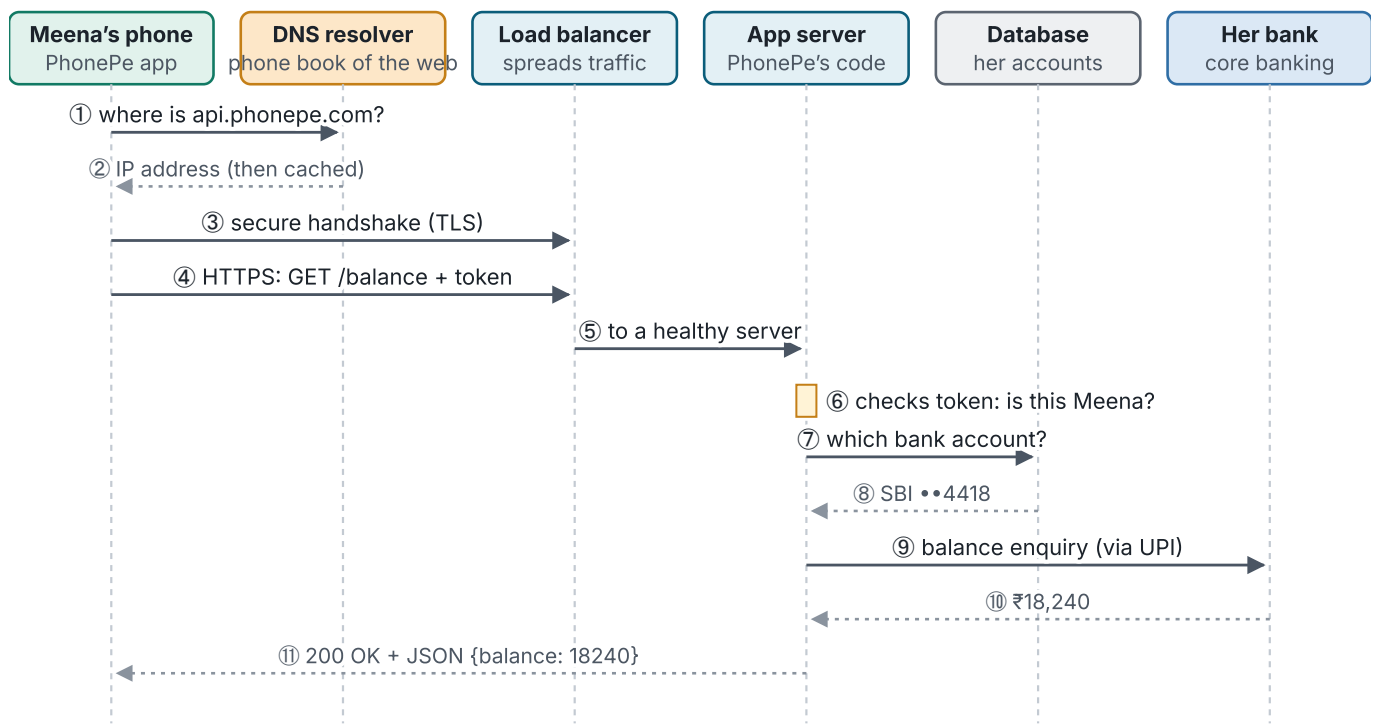
"How do you work with engineers?" "I own the why and the what: the problem, the user story and the acceptance criteria. Engineers own the how. I bring them in at requirements, not after design, because they often see a cheaper way to reach the same outcome." **"The sprint is slipping. What do you do?"** "Protect the date or the scope, not both. I'd rank what's left by user impact, cut from the bottom, and keep testing intact, because in payments a bug costs more than a delay."

System design I: how the internet and APIs work

Meena opens PhonePe on 4G in Indore and taps “Check balance”. Two seconds later she sees ₹18,240. In between, her phone found PhonePe’s servers among billions of machines, set up a secret channel, proved who she is, and PhonePe asked her bank, through NPCI, for a number. A PM who can walk through those two seconds can hold a real conversation about why the app is slow, why a payment is “pending”, and what an engineer means by “the bank’s API is timing out”.

What happens when you open a screen

The app on the phone is the **client**: it asks. Computers in a data centre are the **servers**: they answer. Every screen that shows live data is a conversation between the two, in a fixed order.



Steps ①–⑤ take milliseconds; ⑨–⑩, the bank’s core system, usually take longest.

Figure 1.43 · Behind one “Check balance” tap. Solid arrows are requests, dotted arrows replies. The first steps find and secure the connection; the middle proves who Meena is; the slow part is the bank. *Simplified; real apps reuse connections and cache DNS answers, so steps ①–③ often take no time. Balance illustrative.*

IP address: a server’s numeric address on the internet, like 104.18.2.35.

DNS (domain name system): the internet’s phone book; it turns `api.phonepe.com` into an IP address. Answers are cached, so the lookup is usually skipped.

HTTPS: the web’s request-and-reply format (HTTP), sent inside an encrypted tunnel (TLS). The padlock in a browser means TLS is on.

Load balancer: a traffic police officer that sends each request to a healthy server, so one crash doesn’t take the app down.

Token: a temporary pass the app got at login; every request carries it, so the server knows it is Meena without asking for her password again.

Latency: waiting time. A **round trip** on 4G is often 50–100 ms, so steps that each need a round trip add up.

An API call, opened up

An **API** (application programming interface) is a menu of requests one system accepts from another, with a fixed format for each. **REST** is the most common style: each thing is a web address (`/orders`, `/payments`), and a verb says what to do with it: **GET** to read, **POST** to create. Here is the real shape of the call Neha’s website makes to Razorpay when Arjun taps “Pay ₹899”.

Request: Neha's server → Razorpay

```
POST https://api.razorpay.com/v1/orders
Authorization: Basic <key id : secret>
Content-Type: application/json

{
  "amount": 89900,
  "currency": "INR",
  "receipt": "neha_order_1042"
}
```

- ① the verb and address say what to do
- ② the secret key lives only on Neha's server, never in the app

Response: Razorpay → Neha's server

```
200 OK

{
  "id": "order_Q7xK...",
  "amount": 89900,
  "status": "created"
}
```

- ③ amounts are in paise: 89900 = ₹899, so no rounding errors with decimals

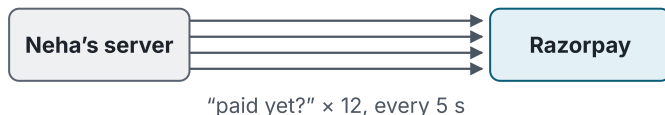
Figure 1.44 · The anatomy of one API call. A request names an action and an object, proves who is asking, and carries data as JSON (labelled text that both sides can read); the reply carries a status code and data. *Format as in Razorpay's public Orders API documentation; values illustrative.*

Status code	Meaning	What the user should see (the PM's call)
200 / 201	OK / created	Success, and the next step
400	Bad request: the data was wrong	A specific fix: "Enter a 10-digit mobile number", never "Error 400"
401 / 403	Not logged in / not allowed	Sign in again, without losing what they typed
429	Too many requests: slow down	Wait quietly and retry; NPCI caps balance checks at 50 per app per user a day
500 / 503 / timeout	The server or a partner failed	For money, say "pending", never "failed", until the status is certain

Webhooks vs polling: how a system hears that something happened

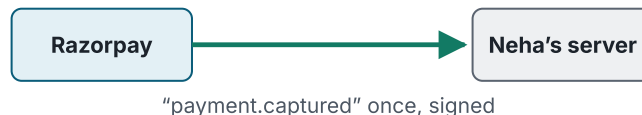
After Arjun pays, Neha's server needs to know. It can **poll**, asking Razorpay every few seconds "paid yet?", like a child asking "are we there yet?". Or Razorpay can send a **webhook**, a message to an address Neha registered in advance: "payment.captured, order_Q7xK, ₹899". The first is simple and wasteful; the second is efficient but needs care.

Polling: ask again and again



10 lakh orders a day × 12 checks
= 1.2 crore calls, mostly "not yet"

Webhook: be told once



10 lakh orders = 10 lakh messages
but: retry if Neha's server is down,
check the signature, ignore duplicates

Figure 1.45 · Polling vs webhooks. Webhooks cut the traffic twelvefold here, but the receiver must handle a message arriving twice or late, and must check the signature so a fake "paid" message can't ship an order. *Illustrative numbers; webhook signing as in Razorpay's public docs (HMAC signature header).*

WHEN IT BREAKS

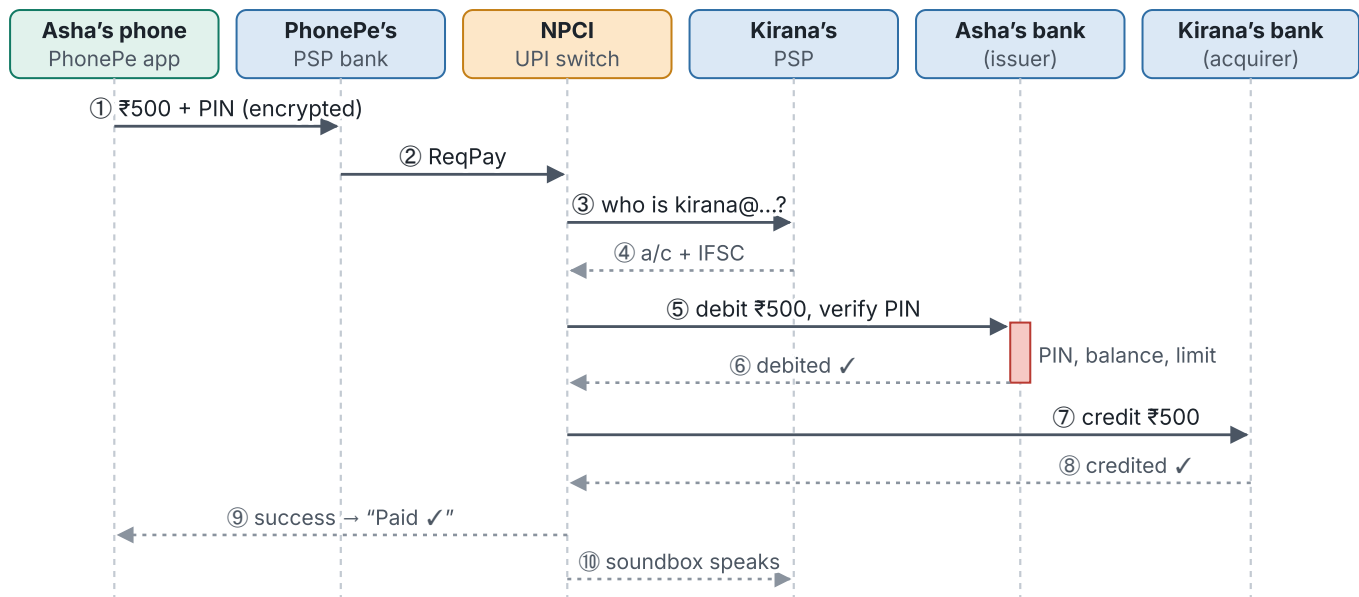
- **A slow partner makes you slow.** If a bank's API takes 8 s, your screen takes 8 s. Timeouts and clear "pending" states matter more than faster code.
- **Retries double-charge** unless each request carries a unique ID that the server remembers (idempotency: a repeated request has the effect of one).
- **Unsigned webhooks** let a fraudster tell a shop "paid" when nothing was paid.

IN THE INTERVIEW

"Our app feels slow. What would you check?" "Where the time goes, step by step: finding the server, the connection, our servers, then partners such as banks. I'd look at p95, not the average, split by partner and network type. Usually one slow partner or a chatty screen making ten calls is the culprit."

Walkthrough: 2 seconds inside a UPI payment

Back at the Pune kirana. Between Asha's thumb on "Pay" and the soundbox speaking, about ten messages cross five organisations, all using the API ideas above. Knowing that path lets a PM answer the question every payments team hears weekly: "The money left my account but the shop didn't get it. Where is it?"



Where ~1.9 s goes (illustrative)

NPCI timeout: 15 s for pay requests, since 16 Jun 2025

①② app + network 350	③④ 200	⑤⑥ issuer bank checks and debits: 800 ms	⑦⑧ credit 300	⑨ reply 250
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typically the slowest hop: the bank's core system (NPCI publishes each bank's decline rate)

Figure 1.46 · A UPI payment as a sequence of messages (time runs downwards). Solid arrows are requests, dotted arrows are replies. Only step ⑤ touches Asha's money, and it happens deep inside her bank, far from the app she is looking at. *Message names from NPCI's UPI specification; timings are illustrative, no public per-hop data exists. Timeout: NPCI circular, 26 Apr 2025.*

ReqPay: UPI's API request for "pay this"; every UPI app, bank and NPCI speak the same message format.

Switch: a router for payment messages. NPCI's switch sees every UPI payment and holds none of the money.

Timeout: how long a system waits before giving up. UPI allows 15 s for pay requests and 10 s for status checks.

Idempotency: sending the same request twice has the effect of sending it once. Each UPI payment carries a unique ID, so a retry can't debit twice.

Pending ("deemed"): debit done, credit not confirmed. A status check or a reversal settles it later.

p95 latency: the time within which 95% of requests finish.

Why the average lies

Take 100 payments: 90 finish in 1.2 s, 8 take 3 s and 2 hang until the 15 s timeout. The mean is $(90 \times 1.2 + 8 \times 3 + 2 \times 15) \div 100 = 1.62$ s, which looks healthy. But the 95th payment in line (**p95**) takes 3 s and the 99th (**p99**) takes 15 s. Users feel their own payment, not the average: the unlucky two stand at the counter staring at a spinner. At 791 mn payments a day, 2 in 100 would be 1.6 crore stuck payments daily (illustrative; real UPI failure rates are lower).



Figure 1.47 · The same 100 payments, three summaries. The mean hides the two payments that hang for 15 s; p95 and p99 show them. Payments teams set targets on p95/p99 and success rate for this reason. *Illustrative numbers.*

Success rates multiply

A payment succeeds only if every hop succeeds, so the hop success rates multiply. Five hops at 99.9% each give $0.999^5 \approx 99.5\%$: one payment in 200 fails, although no single hop looks bad. Let one issuer bank slip to 99.0% and the chain drops to $0.999^4 \times 0.99 \approx 98.6\%$, or one in 72. That is why payments teams track success rate *per bank*, and why an app's success rate can fall with no change to the app.

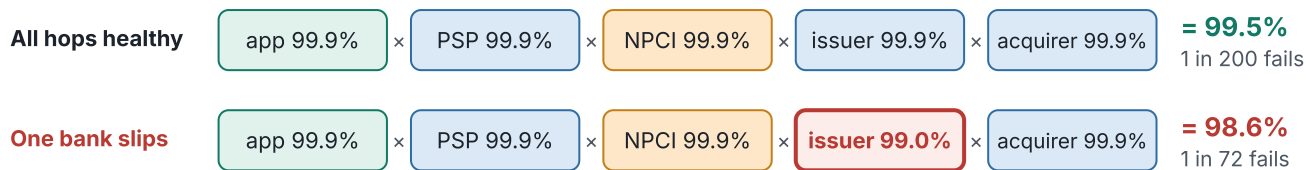


Figure 1.48 · One weak link nearly triples failures. The app, PSP and NPCI did nothing wrong in the second row, yet users see almost three times as many failures and blame the app on their screen. *Illustrative rates; arithmetic exact.*

Uptime	Down per month	Missed*
99.9% (three nines)	43 min	~2.4 crore
99.99%	4.3 min	~24 lakh
99.999%	26 s	~2.4 lakh

*At Aug 2026's average of ~9,150 payments a second (791 mn ÷ 86,400 s). Peaks run higher, so real losses would be larger. Each extra nine cuts downtime tenfold and usually costs far more than ten times as much.

WHEN IT BREAKS · "MONEY GONE, SHOP NOT PAID"

If step ⑤ debits Asha but the reply to ⑦ or ⑧ is lost, the payment is pending. The app must not say "failed" (Asha pays again and is out ₹1,000) or "success" (the kirana hands over goods unpaid). RBI's rule: a failed UPI payment that debited the customer must be reversed by T+1, or the bank pays ₹100 a day.

KEY IDEA · THE PM DECISIONS THIS DRIVES

- What the pending screen says, and how often it re-checks status.
- Retry rules: re-check status freely, never resend a payment with a new ID.
- Routing: PhonePe spreads users across PSP banks (@ybl, @ibl, @axl), so a failing bank can be routed around.
- Dashboards: success rate and p95 per bank, not one average.

IN THE INTERVIEW

"UPI failures are up on our app. Why?" Split by hop before you guess: one issuer bank, one PSP bank, or all of them? Failures clustered on one bank point outside your app. Failures everywhere point at your app, or at NPCI.

A/B testing basics: did the change cause the result?

PhonePe wants more people to pay bills on autopay, because an autopay user comes back every month without being reminded. A designer proposes a button on the success screen: "Autopay next month's bill". The team could ship it to everyone and watch the numbers. But October has Diwali, salary days and a new fee on merchants, so any rise or fall could come from anything. An **A/B test** fixes that: show the button to a random half, hide it from the other half, and compare.

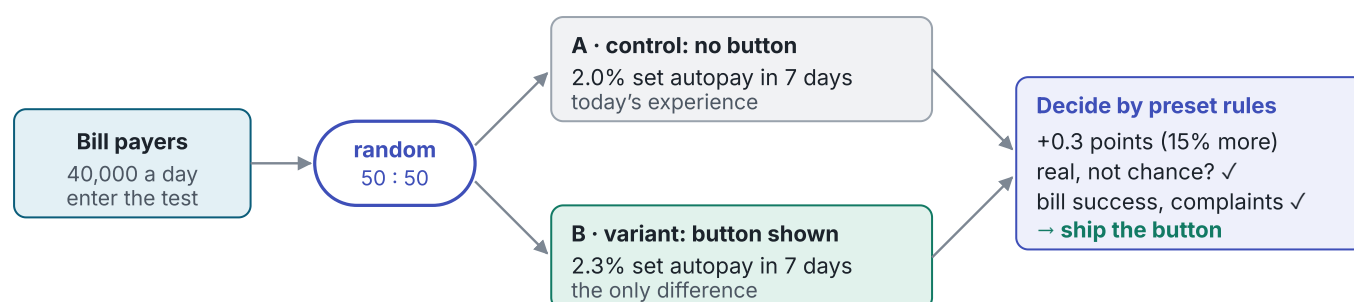


Figure 1.49 · An A/B test in three moves: split at random, change one thing, decide by rules written before the test. Because the split is random, Diwali, salary days and the new fee hit both groups equally, so the gap between them is the button's effect. *Illustrative numbers; method as in IIM A Product Mindset (p. 54).*

Control / variant: the current experience (A) and the changed one (B).

Primary metric: the one number that decides the test, chosen before it starts (autopay set within 7 days).

Guardrail metric: a number that must not get worse (bill-payment success, complaints).

Statistically significant: a gap too large to be luck. By convention, a gap this big would appear by chance less than 5% of the time ($p < 0.05$).

Power: the chance the test spots a real effect of a given size; teams usually aim for 80%.

MDE (minimum detectable effect): the smallest gap worth detecting, which sets the sample size.

Light maths: how many users, how many days?

A handy rule of thumb for an 80%-power test at 5% significance: users needed **in each group** $\approx 16 \times p \times (1 - p) \div \delta^2$, where p is today's rate and δ the gap you want to detect. With $p = 2\%$ and a hoped-for 2.2% ($\delta = 0.2$ points = 0.002): $16 \times 0.02 \times 0.98 \div 0.002^2 = 16 \times 0.0196 \div 0.000004 = 78,400$ **users per group**. At 20,000 new users per group per day, that is 4 days. Halve the gap you want to see and you need **four times** the users, because δ is squared.

Users needed per group, starting from 2.0%

days at 20,000 a day

2.0% → 2.1% (5% lift)	3,13,600	16 days
2.0% → 2.2% (10%)	78,400	4 days
2.0% → 2.4% (20%)	19,600	1 day
2.0% → 3.0% (50%)	3,136	a few hours

Small effects on low rates need big samples; run whole weeks anyway, and include the 1st–10th of the month, when most bills fall due.

Figure 1.50 · Detecting small changes is expensive. Each halving of the effect you want to detect quadruples the users needed. For a small app, a 5% lift on a 2% rate may be untestable, so test bolder changes. *Lehr's rule of thumb (80% power, 5% significance); arithmetic exact, traffic illustrative.*

WHEN IT BREAKS

- **Peeking:** checking daily and stopping on the first good day turns luck into “wins”. Fix the duration in advance.
- **Too many metrics:** test 20 metrics at 5% and about one will look significant by chance. Pick one primary metric.
- **Users who affect each other:** in P2P payments a variant user pays a control user, so the groups leak. Split by city or friend groups instead.
- **Novelty:** a shiny new button gets clicked for a week, then ignored. Run long enough to see it fade.

KEY IDEA · THE PM DECISIONS THIS DRIVES

- The hypothesis and the one primary metric.
- The smallest effect worth shipping, which sets the size and length.
- The guardrails that can veto a “win”.
- What happens if the result is flat: stop, or try a bolder variant.
- When not to test: legal fixes, or changes too small to measure.

IN THE INTERVIEW

“How would you test this feature?” Answer in one breath: “My hypothesis is that the autopay button raises autopay sign-ups. Primary metric: autopay set within 7 days. Split bill payers 50:50 at random. I'd want to detect 2.0% to 2.2%, which needs about 78,000 users per group, so I'd run two full weeks across the bill cycle. Guardrails: bill-payment success and complaints. Ship if the primary metric wins and the guardrails hold.”

Connect the dots

Four ideas ran through every case, product and technology topic here. Seen together, they are most of what a payments PM thinks about in a week.

Thread	In the cases	In the products	In the technology
Free payments, paid products	Teen credit needed a lender and the law before features; the rural bank's win was fewer branch trips, not app downloads	PhonePe and Paytm earn on loans and devices; Groww earns nothing on the SIP that brings users in	Lending partners plug in through APIs; consent and disbursal rules shape every screen
Trust while money is in flight	The "fail-proof" wallet idea broke on risk; fraud rules must count honest users slowed	Soundboxes beat fake "paid" screen-shots; Razorpay keeps buyers' money in escrow	Pending states, timeouts, idem-potent retries and signed webhooks
Measure what you caused	Google Pay coupons: redemptions are not incremental sales without a holdout	Scratch cards funded by partners; Zerodha's profit held while its trading income fell	A/B tests: one primary metric, guardrails, sample size set in advance
Ship without breaking	The Android update that crashed on newer phones	Paytm's regulatory collapse and re-recovery; PhonePe's paused IPO	Agile loops, staged roll-outs, p95 latency and per-bank success rates

BACK TO SURAT · A SKETCH ANSWER FOR HITESH

What changes on 15 October. On ₹25 lakh a month of payments above ₹2,000, Hitesh now pays about ₹10,000 a month. His cheapest escape is asking buyers to pay his personal UPI ID, which is free. That hurts PhonePe twice: its share of the new fee disappears, and business payments become harder to see, which weakens the payment history it would use to lend to him.

Scope first. Goal: keep large merchants' business payments on their business QR. Who: high-ticket merchants taking ₹1 lakh+ a month. Constraint: the fee is set by NPCI; PhonePe can't waive it.

What I'd build. (1) A plain fee statement: the fee per payment, per day and per month, so nobody is surprised. (2) Reasons to stay that are worth more than 0.4%: next-morning settlement, GST-ready invoices for every payment, and working-capital credit priced on his business QR history. (3) A soundbox message that confirms business payments by the buyer's firm name, which a personal-UPI transfer can't.

Goal (North Star): share of large merchants' payment value still on business QRs 60 days after 15 October. **Safety check (guardrail):** large merchants switching to a rival app, and support complaints about fees. **Trade-off:** credit offers bring in revenue but carry the risk of defaults, so start with merchants who have 12 months of history.

KEY IDEA · THE PAYMENTS PM'S THREE QUESTIONS

1. Who pays for this, and who earns?
2. What happens if money is stuck halfway?
3. How will we know our change caused the result?

IN THE INTERVIEW

In any payments case, ask question 1 within the first two minutes. It is the fastest way to show you understand the industry, and it shapes every metric that follows.